
Muon Dynamics as a Spectral Wasserstein Flow

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Abstract

Gradient normalization stabilizes deep-learning optimization, and spectral normalizations are especially natural for matrix-shaped parameter blocks; Muon is the motivating example. We study an idealized deterministic, continuous-time, vanishing-momentum version of this idea in the mean-field regime, where wide models are represented by probability measures on parameter space. Starting from normalized matrix flows, we introduce Spectral Wasserstein distances indexed by norms γ on positive semidefinite matrices: the trace norm gives classical W_2 , the operator norm gives the Muon geometry, and Schatten norms interpolate between them. We develop the static Kantorovich formulation, a max-min robust-cost representation, Gaussian reductions extending the Bures formula, and for monotone norms, prove equivalence with a Benamou–Brenier formulation. This yields a gradient-flow interpretation of the mean-field normalized training dynamics. We illustrate these findings by numerical experiments on MMD flows, Gaussian reductions, two-layer ReLU models, and shallow attention.

1 Introduction

Spectrally normalized optimizers such as Muon replace the Euclidean geometry of gradient descent by a matrix-aware geometry adapted to block-shaped parameters. This paper presents a mathematical framework for these optimizers in the setting of very wide, possibly infinitely wide, layers by introducing a Spectral Wasserstein geometry that models their dynamics as mean-field gradient flows.

Normalized gradient descent and mean field dynamics. On the optimization side, a growing literature studies normalized first-order methods. The recent framework of Pethick et al. [2025] is particularly relevant because it treats norm-constrained linear minimization oracles as a general language for normalized gradient methods and includes spectral normalizations as special cases. Earlier works such as Cutkosky and Mehta [2020] and Murray et al. [2019] show how normalized gradient methods change the optimization dynamics even in nonconvex settings. For deep architectures, matrix-aware normalizations are especially natural: Shampoo is an early influential example of tensor/matrix-aware preconditioning [Gupta et al., 2018], and Muon has become a leading example of spectral normalization in large-scale training [Jordan et al., 2024, Liu et al., 2025]. This connects naturally with the mean-field description of wide neural networks through probability measures on parameter space, which underlies the landscape analysis of two-layer networks by Mei et al. [2018], the optimal-transport convergence analysis of over-parameterized models by Chizat and Bach [2018], and the metric-gradient-flow viewpoint developed in Ambrosio et al. [2008]. Our work keeps this mean-field perspective but changes the underlying metric from Euclidean Wasserstein geometry to matrix-aware Spectral Wasserstein geometries.

Generalized optimal distances. On the transport side, our work is closest in spirit to generalized coupling costs and weak transport, as developed for instance by Gozlan et al. [2017], Backhoff-

Veraguas et al. [2019], and Backhoff-Veraguas and Pammer [2022]. It is also related to covariance-dependent transport costs [Burger et al., 2025], to matrix-valued optimal transport [Ning et al., 2015, Chen et al., 2018], and to the Bures–Wasserstein geometry of covariance matrices [Bhatia et al., 2019]. Another useful perspective is to robustify the Wasserstein distance by optimizing over the cost itself. The closest antecedent is the subspace robust Wasserstein distance of Paty and Cuturi [2019], which corresponds to the special case of our spectral Wasserstein distance W_γ obtained when γ is a Ky Fan norm [Fan, 1951]. Maximizing over costs is also connected to metric learning in Wasserstein discriminant analysis [Flamary et al., 2018], ground metric learning [Cuturi and Avis, 2014], and congestion models for transportation networks [Carlier et al., 2008]. The opposite direction, minimizing over costs, appears for instance, in Sebbouh et al. [2024]; this leads to a concave minimization problem useful for Gromov–Wasserstein-type structure. By contrast, our robustification is a concave maximization over admissible quadratic costs, leading to a convex static problem.

Contributions. We first derive spectral normalized matrix flows and their mean-field continuity equations in Section 2. We then introduce the static Spectral Wasserstein cost in Section 3, prove its comparison with W_2 in Proposition 3.2, its max-min cost-robust representation in Theorem 3.3, and its Gaussian reduction in Proposition 3.4. For monotone norms, Section 4 proves the equivalence between the static and Benamou–Brenier formulations in Theorem 4.2, yielding the metric property in Corollary 4.3 and explicit geodesics in Corollary 4.4. Section 5 identifies the normalized PDE as the corresponding metric gradient flow and discusses MLP, shallow-attention, MMD, and linear networks examples. The code used to reproduce the numerical experiments is available at <https://github.com/gpeyre/muon-dynamics>. **What this paper shows and what it does not.** The goal of this paper is not to decipher why Muon-type algorithms sometimes outperform vanilla gradient descent for specific architectures, in particular, attention layers. It is to lay out the mathematical formalism needed to explore these questions in the setting of very wide networks. This may be a more modest goal, but it also opens the door to many theoretical questions beyond deep-network training, centered on this new class of transportation geometries.

2 Spectral Gradient Flow and Mean Field Dynamics

Spectral gradient flows on matrices. This section introduces the normalized dynamics studied in the paper, whose geometry is governed by a matrix gauge γ . The “muon-type” descent corresponds to the case where γ is the operator norm. We also assume monotonicity, which is used in the static–dynamic equivalence of Theorem 4.2 and is crucial for the spectral Wasserstein gradient-flow interpretation in Section 5 (but the other results do not require this assumption).

Definition 2.1 (Monotone gauge γ and norm $N(V)$). *Throughout the paper, γ denotes the restriction to $\mathbb{S}_+^d$ of a norm on $\mathbb{S}^d$. We assume that this restriction is monotone: whenever $0 \preceq S \preceq S'$, one has $\gamma(S) \leq \gamma(S')$. The matrix norm induced by γ is $N(V) := \gamma(V^\top V)^{1/2}$; for $\gamma = \gamma_p$, one has $N(V) = \|V\|_{S_{2p}}$. As shown in Proposition A.3, $N(V)$ is indeed a norm when γ is monotone.*

The canonical examples used throughout this article are the Schatten gauges: if $\lambda(S)$ denotes the vector of eigenvalues of $S \succeq 0$, then $\gamma_p(S) := \|S\|_{S_p} = \|\lambda(S)\|_{\ell_p}$, $p \in [1, \infty]$. Other examples include the Ky Fan norms [Fan, 1951], $\gamma(S) := \sum_{i=1}^k \lambda_i(S)$, the sum of the k largest eigenvalues, which interpolate between the operator norm ($k = 1$) and the trace norm ($k = d$), which is exactly the setting considered by Paty and Cuturi [2019] for subspace robust Wasserstein distance.

Definition 2.2 (Linear Minimization Oracle (LMO)). *For $G \in \mathbb{R}^{n \times d}$, the matrix LMO associated with γ is any minimizer $J_\gamma(G) \in \arg\min_V \{ \langle G, V \rangle_F + \frac{1}{2} N(V)^2 \}$.*

Let $F : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}$ be a smooth objective. Following Pethick et al. [2025], it yields both the discrete step and the ODE flow

$$X_{k+1} = X_k + \tau J_\gamma(\nabla F(X_k)) \quad \text{and} \quad \dot{X}_t = J_\gamma(\nabla F(X_t)). \quad (1)$$

For $p = 1$, $J_{\gamma_1}(G) = -G$, so one recovers the usual gradient flow; for $p = \infty$, J_{γ_∞} is the trace-scaled polar projection of G onto the orthogonal factor, giving a simplified “Muon” ODE. Explicit Schatten formulas are recalled in Appendices A.

85 **Mean-field lift and normalized continuity equation.** Given particles $x_1, \dots, x_n \in \mathbb{R}^d$, write
86 $X := [x_1^\top; \dots; x_n^\top] \in \mathbb{R}^{n \times d}$ and $\mu_X := n^{-1} \sum_i \delta_{x_i}$. We consider permutation-invariant objectives
87 lifted from measures, $F(X) := n\mathcal{F}(\mu_X)$; this is in particular the case for neural-network layers such
88 as two-layer MLPs and attention layers, as exposed in Section 5.

89 **Definition 2.3** (Measure-LMO). *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$. For $v \in L^2(\mu; \mathbb{R}^d)$, define $\mathcal{N}_\mu(v)^2 :=$
90 $\gamma(\int v(x)v(x)^\top d\mu(x))$. For $g \in L^2(\mu; \mathbb{R}^d)$, the measure-LMO is any member of $\mathcal{J}_\gamma^\mu(g) \in$
91 $\operatorname{argmin}_{v \in L^2(\mu; \mathbb{R}^d)} \{ \int g(x) \cdot v(x) d\mu(x) + \frac{1}{2} \mathcal{N}_\mu(v)^2 \}$.*

92 For an empirical measure $\mu_X = n^{-1} \sum_i \delta_{x_i}$ transported by particle velocities stacked into a matrix
93 V , one has $\mathcal{N}_{\mu_X}(v)^2 = n^{-1} \mathbf{N}(V)^2$. If G denotes the matrix with rows $g(x_i)^\top$, then the rows of
94 $\mathcal{J}_\gamma^{\mu_X}(g)$ coincide with those of $\mathcal{J}_\gamma(G)$. For a more general measure μ , however, it is not a priori clear
95 how to compute $\mathcal{J}_\gamma^\mu(g)$. The following theorem shows how to do so through a dual representation.

96 **Definition 2.4** (Representing set $\mathcal{K}_\gamma$). *A representing set for γ is a convex compact set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$
97 such that, for every $S \succeq 0$, $\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \operatorname{tr}(QS)$. As recalled in Appendix A, monotonicity of
98 γ on $\mathbb{S}_+^d$ allows the representing set to be chosen inside $\mathbb{S}_+^d$. Remark A.4 gives examples of such sets
99 $\mathcal{K}_\gamma$ for the Schatten norms $\gamma = \gamma_p$.*

100 **Theorem 2.5** (Structure theorem for the Measure-LMO). *Define $\mathcal{S}_\mu(g) := \int g(x)g(x)^\top d\mu(x)$.
101 Assume that $Q_\mu^* \in \operatorname{argmin}_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \operatorname{tr}(Q^{-1}\mathcal{S}_\mu(g))$ admits a minimizer. Then $\mathcal{J}_\gamma^\mu(g)(x) =$
102 $-(Q_\mu^*)^{-1}g(x)$ is a valid Measure-LMO.*

103 *Proof sketch.* The detailed proof is in Appendix A. Set $S := \mathcal{S}_\mu(g)$. The Measure-LMO problem
104 reduces pointwise to minimizing $\int \langle g(x), v(x) \rangle d\mu(x)$ over the unit ball induced by the dual action.
105 Using the matrix representation from Appendix A, this constraint can be written through the family
106 of quadratic bounds indexed by $Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d$. For fixed Q , the Cauchy–Schwarz inequality
107 in the Q -metric gives $\int \langle g, v \rangle d\mu \geq -(\int v^\top Q v d\mu)^{1/2} (\operatorname{tr}(Q^{-1}S))^{1/2}$, with equality when v is
108 proportional to $-Q^{-1}g$. Optimizing over admissible matrices therefore selects a minimizer $Q_\mu^* \in$
109 $\operatorname{argmin}_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \operatorname{tr}(Q^{-1}S)$. With this choice, the equality case in Cauchy–Schwarz is realized by
110 $v(x) = -(Q_\mu^*)^{-1}g(x)$. The matrix optimality condition from Appendix A ensures that this vector
111 field lies on the appropriate unit boundary, so it attains the minimum in the Measure-LMO problem.
112 Hence $\mathcal{J}_\gamma^\mu(g)(x) = -(Q_\mu^*)^{-1}g(x)$ is a valid selector. $\square$

113 More generally, if the inverse-trace problem admits no minimizer in $\mathbb{S}_{++}^d$, as may occur for $p = \infty$
114 when the optimal matrix lies on the boundary of the positive semidefinite cone, a Measure-LMO is
115 obtained by taking the limit of the same expression along any minimizing sequence in $\mathbb{S}_{++}^d$.

116 Following [Ambrosio et al., 2008], we define the Wasserstein gradient of $\mathcal{F}$ as the vector field
117 $g_\mu(x) := \nabla_x \delta \mathcal{F} / \delta \mu(x)$ where the first variation $\delta \mathcal{F} / \delta \mu$ is defined by $\frac{d}{d\varepsilon} \mathcal{F}(\mu + \varepsilon \sigma)|_{\varepsilon=0} =$
118 $\int (\delta \mathcal{F} / \delta \mu) d\sigma$ for signed zero-mass perturbations. The normalized mean-field dynamics is then
119 the PDE

$$\partial_t \mu_t + \operatorname{div}(\mu_t \mathcal{J}_\gamma^{\mu_t}(g_{\mu_t})) = 0. \quad (2)$$

120 When $\gamma(S) = \operatorname{tr}(S)$, $\mathcal{J}_\gamma^\mu(g) = -g$ and this reduces to the classical W_2 gradient-flow PDE. For
121 discrete measures, one recovers the ODE (1).

122 **Proposition 2.6** (Finite-width interpretation). *Assume the normalized continuity equation starting
123 from μ_{X_0} preserves empirical measures and can be written as $\mu_t = n^{-1} \sum_i \delta_{x_i(t)}$, denote $X_t :=$
124 $[x_1(t)^\top; \dots; x_n(t)^\top]$. Then X_t solves the ODE (1) if and only if μ_{X_t} solves (2) weakly. The proof is
125 in Appendix A.*

126 Extending the global-in-time PDE well-posedness theory of Chizat and Bach [2018], which covers
127 the trace case $p = 1$ under smoothness and growth assumptions on the feature map, to general
128 Schatten geometries is a difficult open problem left for future work. The major question left open
129 by the construction above is whether (2) is the gradient flow of an actual distance on probability
130 measures. The next two sections answer this question: first by defining a static Spectral Wasserstein
131 cost, and then by proving that, for monotone norms, this cost admits a Benamou–Brenier dynamic
132 formulation whose infinitesimal norm is precisely $\mathcal{N}_\mu$.

Remark 2.7 (Connection with Muon). *Appendix B relates the normalized-flow viewpoint to two practical ingredients of Muon. Finitely many Newton–Schulz iterations replace the exact LMO by a polynomial spectral map generated by an explicit functional γ . Momentum can be modeled by an exponentially averaged force variable, leading in the mean-field limit to a phase-space Liouville equation; this is no longer a gradient flow on the position marginal alone, and thus lies partly outside the Spectral Wasserstein geometry developed below.*

3 Static Spectral Wasserstein Geometry

Kantorovich cost. This section introduces the Spectral Wasserstein transport cost. The key point is that the correct object is coupling-based and depends on a norm γ acting on the global displacement covariance.

Definition 3.1 (Static Spectral Wasserstein). *For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, let $\Pi(\mu, \nu)$ be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν , and define*

$$W_\gamma(\mu, \nu)^2 := \inf_{\pi \in \Pi(\mu, \nu)} \gamma \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)(y - x)^\top d\pi(x, y) \right).$$

The Monge restriction, the possible strictness of the coupling relaxation, and the obstruction to the triangle inequality for non-monotone norms are discussed in Appendix C. In the monotone case, the full metric property is proved later through the dynamic formulation in Corollary 4.3.

Comparison with W_2 and cost-robust formulation. We now show that W_γ has the same topology as W_2 . The proof as well as explicit Schatten constants and sharpness examples, are given in Appendix A.2.

Proposition 3.2 (Norm comparison). *Let $0 < c_\gamma \leq C_\gamma < \infty$ such that $c_\gamma \operatorname{tr}(S) \leq \gamma(S) \leq C_\gamma \operatorname{tr}(S)$ for every $S \succeq 0$ (such constant always exist). Then $\sqrt{c_\gamma} W_2(\mu, \nu) \leq W_\gamma(\mu, \nu) \leq \sqrt{C_\gamma} W_2(\mu, \nu)$.*

The comparison bounds of Proposition 3.2 already imply separation and topological equivalence with W_2 . What is not immediate from the static formulation alone is the triangle inequality. To address this, and to clarify the geometry of W_γ , Theorem 3.3 below recasts W_γ as a cost-robust Wasserstein distance. This reformulation has two immediate consequences: a conditional Brenier theorem in the Monge setting (Appendix C) and the fact that W_γ is a distance (Appendix A.3). The latter can also be obtained from the Benamou–Brenier formulation; see Corollary 4.3.

Theorem 3.3 (Max-min and cost-robust representation). *For every symmetric matrix $Q \in \mathbb{S}^d$, let W^Q denote the quadratic transport functional associated with the cost $\langle Q(x - y), x - y \rangle$, namely $W^Q(\mu, \nu)^2 := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)^\top Q(y - x) d\pi(x, y)$. In particular, $W^{\operatorname{Id}} = W_2$. Then, denoting $\mathcal{K}_\gamma$ any representing set for γ as in Definition 2.1, for every $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} \inf_{\pi \in \Pi(\mu, \nu)} \int (y - x)^\top Q(y - x) d\pi(x, y) = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu)^2.$$

Denoting $\Sigma_\pi := \int (y - x)(y - x)^\top d\pi(x, y)$, any saddle point (Q^*, π^*) satisfies $Q^* \in \partial\gamma(\Sigma_{\pi^*})$.

Proof sketch. Details are given in Appendix A.2. The support representation of γ gives $\gamma(\Sigma_\pi) = \max_{Q \in \mathcal{K}_\gamma} \operatorname{tr}(Q\Sigma_\pi)$, with $\operatorname{tr}(Q\Sigma_\pi) = \int (y - x)^\top Q(y - x) d\pi(x, y)$. Hence $W_\gamma(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \max_{Q \in \mathcal{K}_\gamma} F(\pi, Q)$, where $F(\pi, Q) := \int (y - x)^\top Q(y - x) d\pi(x, y)$. The conclusion follows by exchanging the infimum and the maximum. Indeed, $\Pi(\mu, \nu)$ is convex, $\mathcal{K}_\gamma$ is compact and convex, and F is affine in both variables, continuous in Q , and lower semicontinuous in π . Sion’s minimax theorem therefore, gives $\inf_\pi \max_Q F(\pi, Q) = \max_Q \inf_\pi F(\pi, Q)$. The inner infimum is precisely $W^Q(\mu, \nu)^2$, yielding $W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu)^2$. $\square$

Discrete setting. This subsection spells out the finite-dimensional version used in computations and shows how the abstract coupling problem becomes a concrete convex program. We consider two discrete measures $\mu := \sum_{i=1}^n a_i \delta_{x_i}$ and $\nu := \sum_{j=1}^m b_j \delta_{y_j}$, with weights $a_i, b_j \geq 0$ summing to one. As in classical Kantorovich transport, a coupling is represented by a transport matrix $P \in \mathbb{R}_+^{n \times m}$

satisfying $P\mathbf{1}_m = a$ and $P^\top \mathbf{1}_n = b$. The discrete static Spectral Wasserstein problem can then be written as the convex optimization problem

$$W_\gamma(\mu, \nu)^2 = \min_{P \geq 0, P\mathbf{1}_m = a, P^\top \mathbf{1}_n = b} \gamma \left(\sum_{i=1}^n \sum_{j=1}^m P_{ij} (y_j - x_i)(y_j - x_i)^\top \right).$$

For $\gamma = \gamma_p$ and $p = 1$ this reduces to the usual linear optimal-transport problem with quadratic cost, hence to a classical assignment problem in the equal-weight case. For $p = \infty$, the problem reduces to a positive-semidefinite programming problem on the displacement covariance. Figure 1 shows how the optimal matching changes from the trace to the operator endpoint. Proposition A.8 explains this effect in the regime where the two measures have well-separated means: while the trace case remains isotropic, every Schatten geometry with $p > 1$ asymptotically selects a quadratic cost aligned with the mean displacement direction, making the matching primarily sensitive to displacements parallel to the vector joining the two means. Note that, as explained in Appendix C, for $p = +\infty$ one does not expect the existence of a Monge map even for measures with density since the optimal Q^* is in general rank 1 (see discussion in Remark 5.2) and this is exactly what is hinted by Figure 1. Additional image-defined shape pairs are shown in Appendix G.

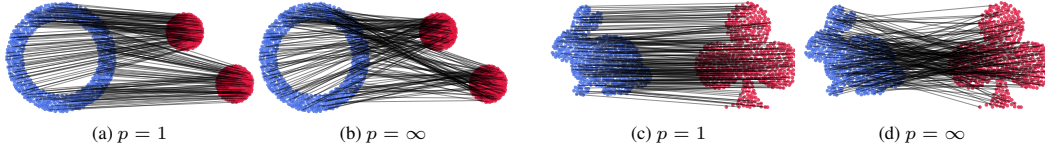


Figure 1: Static spectral couplings for two source-target pairs. Each pair is shown for $p = 1$ and $p = \infty$, with only a subsample of matched segments displayed.

Gaussian setting and Spectral Bures distance. Gaussian marginals compress the transport problem to a finite-dimensional optimization over admissible covariance blocks. The proof of the following proposition 3.4 and the closed commuting formula for Schatten norms are collected in Appendix D.

Proposition 3.4 (Gaussian reduction). *Let $\mu = \mathcal{N}(m_0, \Sigma_0)$ and $\nu = \mathcal{N}(m_1, \Sigma_1)$. Then $W_\gamma(\mu, \nu)^2 = \inf_K \left\{ \gamma((m_1 - m_0)(m_1 - m_0)^\top + \Sigma_0 + \Sigma_1 - K - K^\top : \begin{pmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{pmatrix} \succeq 0) \right\}$.*

4 Dynamic Formulation and Geodesics

We now turn to the Benamou–Brenier side [Benamou and Brenier, 2000]. This section assumes that γ is monotone, so Appendix A allows us to choose $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$; this positive-semidefinite reduction is the key point in the proof of Theorem 4.2, stating the dynamic-static equivalence.

Definition 4.1 (Dynamic Spectral Wasserstein). *For $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$, define $W_\gamma^{\text{BB}}(\mu_0, \mu_1)^2$ as the infimum of $\int_0^1 \gamma(\int v_t(x)v_t(x)^\top d\mu_t(x))dt$ over narrowly continuous curves and velocities satisfying $\partial_t \mu_t + \text{div}(\mu_t v_t) = 0$ with endpoints μ_0, μ_1 .*

The equivalent convex momentum formulation is given in Appendix E. The following theorem, which is the main result of the paper, states the equivalence between the static and dynamic formulations. The proof of the theorem and its corollaries are in Appendix E.

Theorem 4.2 (Static-dynamic equivalence). *If γ is monotone, one has $W_\gamma^{\text{BB}}(\mu_0, \mu_1) = W_\gamma(\mu_0, \mu_1)$.*

Proof sketch. The detailed argument is given in Appendix E. For the dynamic-to-static inequality, let (μ_t, v_t) solve the continuity equation. By the superposition principle, it is represented by a probability measure η on absolutely continuous curves, with $\gamma_{t\#}\eta = \mu_t$ and $\dot{\gamma}_t = v_t(\gamma_t)$ for a.e. t . The endpoint coupling $\pi = (\gamma_0, \gamma_1)_\# \eta$ then satisfies $\gamma_1 - \gamma_0 = \int_0^1 v_t(\gamma_t) dt$ η -a.e. By monotonicity and Appendix A, the dual support set $\mathcal{K}_\gamma$ may be chosen in $\mathbb{S}_+^d$. Hence, for every $Q \in \mathcal{K}_\gamma$, Jensen’s inequality gives $\langle Q, (\gamma_1 - \gamma_0)(\gamma_1 - \gamma_0)^\top \rangle \leq \int_0^1 \langle Q, v_t(\gamma_t)v_t(\gamma_t)^\top \rangle dt$. Integrating with respect to η and taking the supremum over $Q \in \mathcal{K}_\gamma$ yields the static cost of π bounded by the dynamic action. Taking the infimum over admissible (μ_t, v_t) gives one inequality. Conversely, let π be an optimal static coupling and set $\gamma_t = (1 - t)x + ty$ for $(x, y) \sim \pi$. Then $\dot{\gamma}_t = y - x$, and the induced curve

201 solves the continuity equation. Its dynamic action equals the static cost of π , giving the reverse
 202 inequality. $\square$

203 **Corollary 4.3** (Metric properties). W_γ is a distance on $\mathcal{P}_2(\mathbb{R}^d)$ inducing the same topology as W_2 .

204 **Corollary 4.4** (Geodesics). If π_* is any optimal coupling for $W_\gamma(\mu_0, \mu_1)$, then $\mu_t = ((1-t)x +$
 205 $ty)_{\#}\pi_*$ is a constant-speed geodesic and $W_\gamma(\mu_s, \mu_t) = |t-s|W_\gamma(\mu_0, \mu_1)$ for every $0 \leq s \leq t \leq 1$.

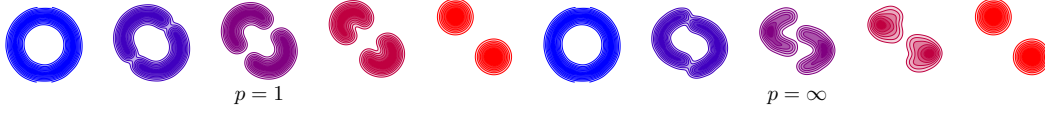


Figure 2: Displacement-interpolation density snapshots induced by optimal static couplings, for $p = 1$ (first five panels) and for $p = \infty$ (last five panels). We use $n = 1500$ particles. As in Figure 1, the first and last snapshots correspond to the same source and target distributions ($t = 0$ and $t = 1$).

206 Figure 2 complements Figure 1: once an optimal static coupling is fixed, the induced displacement
 207 interpolation remains linear in time at the particle level, but the macroscopic density evolution differs
 208 markedly between $p = 1$ and $p = \infty$. Appendix G shows additional examples.

209 5 Spectral Wasserstein Gradient Flows

210 We now return to the normalized continuity equation introduced in Section 2 and explain why, after
 211 the static and dynamic theory above, it deserves to be called a Spectral Wasserstein gradient flow.

212 **Metric interpretation of the normalized PDE.** We now show that (2) is the gradient flow of $\mathcal{F}$ for
 213 the metric W_γ , in the metric-gradient-flow sense of Ambrosio et al. [2008]. Recall that the metric
 214 derivative of an absolutely continuous curve $t \mapsto \mu_t$ is $|\mu'|_\gamma(t) := \lim_{h \rightarrow 0} W_\gamma(\mu_{t+h}, \mu_t)/|h|$, when-
 215 ever the limit exists, and the local slope of $\mathcal{F}$ is $|\partial\mathcal{F}|_\gamma(\mu) := \limsup_{\nu \rightarrow \mu} (\mathcal{F}(\mu) - \mathcal{F}(\nu))_+/W_\gamma(\mu, \nu)$.
 216 A W_γ -gradient flow of $\mathcal{F}$ is a curve of maximal slope, i.e., a curve satisfying the energy dissipation
 217 identity $\frac{d}{dt}\mathcal{F}(\mu_t) = -\frac{1}{2}|\mu'|_\gamma(t)^2 - \frac{1}{2}|\partial\mathcal{F}|_\gamma(\mu_t)^2$.

218 **Theorem 5.1** (W_γ gradient flow). If γ is monotone and $\mathcal{F}$ is sufficiently smooth, then any sufficiently
 219 regular solution of (2) is a curve of maximal slope for $\mathcal{F}$ with respect to W_γ .

220 *Proof sketch.* The detailed proof is in Appendix E.4. The dynamic formulation identifies the tangent
 221 norm associated with W_γ as the action density $\gamma(\int v v^\top d\mu)^{1/2}$. Therefore, for a smooth curve (μ_t, v_t)
 222 solving the continuity equation, the metric speed satisfies $|\dot{\mu}_t|_\gamma^2 = \gamma(\int v_t v_t^\top d\mu_t)$. Along such a
 223 curve, the first variation formula gives $\frac{d}{dt}\mathcal{F}(\mu_t) = \int \langle \nabla_{\delta\mu} \mathcal{F}(\mu_t), v_t \rangle d\mu_t$. The duality formula for the
 224 action norm then yields $\frac{d}{dt}\mathcal{F}(\mu_t) \geq -|\partial\mathcal{F}|_\gamma(\mu_t) |\dot{\mu}_t|_\gamma$. For the vector field prescribed by (2), equality
 225 holds in this duality bound: the velocity is the Measure-LMO applied to $\nabla_{\delta\mu} \mathcal{F}(\mu_t)$, hence it realizes
 226 the steepest descent direction for the W_γ tangent norm. Consequently, $-\frac{d}{dt}\mathcal{F}(\mu_t) = |\partial\mathcal{F}|_\gamma(\mu_t) |\dot{\mu}_t|_\gamma$,
 227 and the normalization in (2) gives $|\dot{\mu}_t|_\gamma = |\partial\mathcal{F}|_\gamma(\mu_t)$. Thus, the energy dissipation identity holds:
 228 $\frac{d}{dt}\mathcal{F}(\mu_t) = -|\dot{\mu}_t|_\gamma^2 = -|\partial\mathcal{F}|_\gamma(\mu_t)^2$. This is precisely the curve-of-maximal-slope condition for $\mathcal{F}$
 229 with respect to W_γ . $\square$

230 **Remark 5.2** (Insight on Muon's dynamics). In the “Muon setting”, corresponding to the Schatten- ∞
 231 gauge $\gamma(S) = \lambda_{\max}(S)$, the optimality relation $Q^* \in \partial\gamma(\Sigma_{\pi^*})$ stated in Theorem 3.3 implies that
 232 Q^* is supported on the top eigenspace of the optimal displacement covariance Σ_{π^*} . If this leading
 233 eigenspace is one-dimensional (which is the generic situation), say $\text{span}(u^*)$, then $Q^* = u^*(u^*)^\top$
 234 and π^* is optimal for the projected quadratic cost $\langle y - x, u^* \rangle^2$. Thus, locally, the Muon mean-field
 235 dynamics can be viewed as a Wasserstein gradient flow in which the relevant ground cost is not the
 236 full Euclidean quadratic cost, but a one-dimensional projected cost selected by the current largest
 237 displacement or velocity covariance direction. In this sense, Muon does not uniformly penalize all
 238 directions of motion. Instead, its geometry normalizes the dominant direction: the active metric
 239 looks only at the direction where the covariance of the infinitesimal displacement is largest, and
 240 the resulting steepest descent field is obtained by the corresponding projected optimal-transport
 241 geometry. This provides a geometric interpretation of Muon's orthogonalizing effect: the dynamics

continuously identifies the dominant direction in the mean-field motion and rescales the flow through this worst-direction cost.

Mean-field models for two-layer MLPs. We now specialize the abstract gradient-flow PDE to the standard mean-field parameterization of two-layer MLPs. The mean-field predictor is $H_\mu(z) := \int_\Omega \phi(z, x) d\mu(x)$, and the training objective is $\mathcal{F}(\mu) := R(H_\mu)$ for some loss R . For a two-layer network with scalar output and activation σ , one may take parameter $x = (u, v) \in \mathbb{R}^d$ and feature $\phi(z, x) := u \sigma(v \cdot z)$. This is the standard mean-field representation of a two-layer multilayer perceptron [Mei et al., 2018, Chizat and Bach, 2018]. The notation is a deliberate departure from usual machine-learning conventions: the network input is denoted by z , while the trainable variable is denoted by x so that it matches the transport notation used throughout the paper. Here d is the sum of the block dimensions of u and v . In practical implementations of Muon one often normalizes separate parameter blocks, whereas in our continuum model the whole particle x is treated as a single vector; this is the natural mean-field analogue of a single normalized block. A basic example is the quadratic risk $R(H) = \frac{1}{2} \int |H(z) - y^*(z)|^2 d\rho(z)$, or its empirical version on a dataset. Because H_μ depends linearly on μ , this makes $\mathcal{F}(\mu)$ a quadratic interaction functional. These energies are of MMD-type [Gretton et al., 2012], which is why the MMD experiment below is a natural test case for the Spectral Wasserstein flow. In Figure 3, we train a two-dimensional to one-dimensional ReLU network, with $\sigma(s) = \max(s, 0)$, particles $x = (u, v_1, v_2) \in \mathbb{R}^3$, and covariates $z \in \mathbb{R}^2$. The target y^* is generated by a three-neuron teacher on Gaussian covariates $z \sim \mathcal{N}(0, \text{Id}_2)$. To compare the training dynamics for $p = 1$ and $p = \infty$ on the same scale, we rescale time so that the two flows have comparable small-time energy decay. We display the particle evolution in the reduced two-dimensional plane $|u|v$. Global convergence corresponds to the concentration of particles toward the rays associated with the teacher neurons, shown as dashed half-lines in Figure 3(c)–(d), and can also be visualized by the convergence of the circular histogram toward three Dirac masses. Changing p has a strong effect on the early convergence profile, with $p = \infty$ reaching a significantly lower energy level during the initial phase. The ReLU case, and more generally, positively two-homogeneous ϕ admits a spherical reduction leading to a generalized unbalanced transport geometry; this is described in Appendix I.

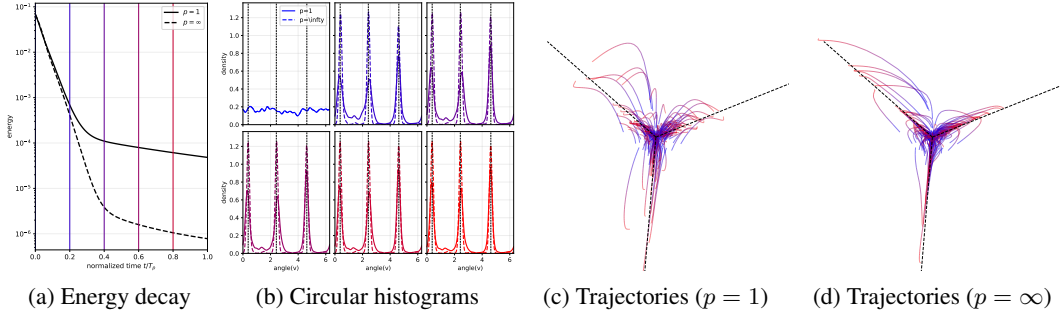


Figure 3: Two-layer ReLU MLP training under spectral flows. (a) Energy decay in normalized time (solid: $p = 1$, dashed: $p = \infty$), with colored vertical bars marking the snapshot times used in (b). (b) Circular histogram snapshots at matched normalized times. (c)–(d) Particle trajectories in the reduced parameter plane ($|u|v_1, |u|v_2$) for $p = 1$ and $p = \infty$.

Mean-field shallow attention. Using the same spectral-flow construction, we also consider a shallow mean-field multi-head attention model. This is a useful stress test because attention uses softmax nonlinearities, precisely the regime where Muon-type spectral normalization is empirically attractive. A datum is a sequence $z = (z_1, \dots, z_m)$ of tokens in $\mathbb{R}^{d_T}$. A head is a particle $x = (Q, K, V, O)$, with $Q, K, V, O \in \mathbb{R}^{d_H \times d_T}$, so $x \in \mathbb{R}^d$ with $d = 4d_H d_T$. Its elementary feature is the softmax attention map $\phi(z, x)_j := O^\top \sum_{i=1}^m \frac{\exp(\langle Qz_j, Kz_i \rangle / \sqrt{d_H})}{\sum_{\ell=1}^m \exp(\langle Qz_j, Kz_\ell \rangle / \sqrt{d_H})} V z_i$, and the mean-field layer is again $H_\mu(z) = \int \phi(z, x) d\mu(x)$. For an empirical measure μ_X , this is a multi-head attention layer averaged over an arbitrary number n of heads, which is the natural mean-field analogue of increasing the number of heads without fixing a finite architecture. Figure 4 reports an experiment with $m = 16$ tokens, $d_T = 4$, and $d_H = 3$: we fit a random five-head teacher by least squares using a student with $n = 200$ heads, and compare the trace and operator geometries over five random initializations. The query/key scale controls the sparsity of the teacher attention matrix. We quantify sparsity through

the effective number of attended tokens $N_{\text{eff}} := (\sum_i w_i^2)^{-1}$ averaged over attention rows: the dense high-temperature regime has $N_{\text{eff}} \approx 16$ out of $m = 16$, while the sparse low-temperature regime has $N_{\text{eff}} \approx 2.1$. The $p = 1$ and $p = \infty$ horizons are rescaled so that small-time energy decay rates are comparable; even after this normalization, changing p strongly affects the convergence profile and $p = \infty$ reaches a lower energy during the first phase.

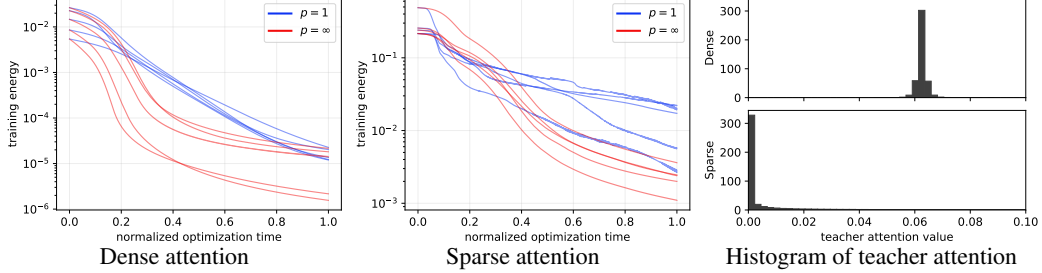


Figure 4: Shallow-attention training under spectral flows ($p = 1$ in blue, $p = \infty$ in red), with five runs per setting. Dense attention corresponds to high temperature and effective support near all tokens ($N_{\text{eff}} \approx 16$), while sparse attention corresponds to low temperature and effective support around two tokens ($N_{\text{eff}} \approx 2.1$). The right panel stacks the pooled teacher-attention value histograms for the dense and sparse regimes.

MMD gradient flows. As a simple model for generative fitting and sampling, in the spirit of MMD-based generative models [Li et al., 2015, Bińkowski et al., 2018], we also minimize $\mathcal{F}(\mu) = \text{MMD}(\mu, \nu)^2$ for the energy-distance kernel $k(x, y) = -\|x - y\|_2$. For empirical measures $\mu = n^{-1} \sum_i \delta_{x_i}$ and $\nu = m^{-1} \sum_j \delta_{y_j}$, with $n = m = 200$ particles in $\mathbb{R}^2$, this objective is $\text{MMD}(\mu, \nu)^2 = n^{-2} \sum_{i,i'} k(x_i, x_{i'}) + m^{-2} \sum_{j,j'} k(y_j, y_{j'}) - 2(nm)^{-1} \sum_{i,j} k(x_i, y_j)$, with a small smoothing of $\|x - y\|_2$ in the implementation. The same initial cloud evolves with $p = 1$ and $p = \infty$. Figure 5 isolates the effect of the spectral tangent norm without the additional architectural structure of a neural network. The trace-norm flow reacts locally to the MMD force field, while the operator-norm flow produces more globally coordinated trajectories; nevertheless, the scalar loss curves remain close.

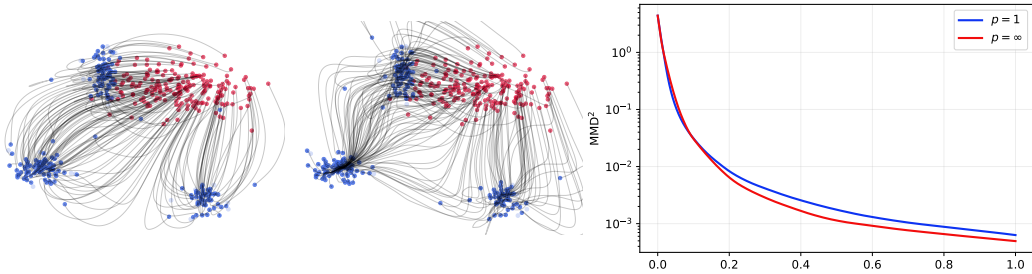


Figure 5: MMD flow trajectories and objective decay for $p = 1$ and $p = \infty$. The left and middle panels show all particle trajectories, while the right panel shows the decay of $\text{MMD}(\mu_t, \nu)^2$.

Gaussian-preserving linear networks. To gain a better understanding of spectrally normalized dynamics, a tractable special case is obtained by taking a linear two-layer network, i.e. $\sigma = \text{Id}$, with particles $x = (u, v)$ and predictor $H_\mu(z) = \int (u^\top z) v d\mu(u, v)$. For the least-squares risk $R(H) = \frac{1}{2} \int |H(z) - y^*(z)|^2 d\rho(z)$, the objective depends only on the first two moments of μ . Write $C_\rho := \int z z^\top d\rho(z)$ for the data covariance and $B_\rho := \int y^*(z) z^\top d\rho(z)$ for the teacher cross-moment. As explained in Appendix F, Gaussian measures are then preserved by the spectral flow, so the infinite-dimensional PDE reduces to an ODE on the mean and covariance. This covariance ODE is still complicated; under simultaneous diagonalization, white-noise balanced initialization, $C_\rho = \text{Id}$, and $B_\rho = \text{diag}(\beta_i)$, it reduces to coupled scalar modes. Writing the modal covariance block as $\Sigma_i(t) = \begin{pmatrix} s_i(t) & r_i(t) \\ r_i(t) & s_i(t) \end{pmatrix}$, the coefficient $r_i(t)$ is the learned predictor coefficient in mode i , while $s_i(t)$ is the corresponding balanced layer variance.

307 **Proposition 5.3** (Commuting Gaussian modal ODE). *Let $q := 2p/(2p - 1)$, then for each mode i ,*
308 *$(\dot{r}_i, \dot{s}_i) = -\mathcal{N}(r, s)^{2-q}(r_i - \beta_i)|r_i - \beta_i|^{q-2}(h_i^+, h_i^-)$, where $h_i^\pm := (s_i + r_i)^{q/2} \pm (s_i - r_i)^{q/2}$ and*
309 *$\mathcal{N}(r, s)^q := \sum_j |r_j - \beta_j|^q h_j^+$.*

310 Note that the coupling factor $\mathcal{N}(r, s)$ is scalar, so the features r_i remain decoupled up to a
311 reparametrization of time and can be analyzed independently. For a rank-one target, the admis-
312 sible state set is the cone $s > |r|$. The predictor coefficient r_t moves toward β , while the terminal
313 value of s_t records the implicit covariance bias selected by p . Figure 6(a) overlays several such
314 trajectories from two initial points: all reach the same predictor line $r = \beta$, but at different covariance
315 levels, revealing a nontrivial p -dependent implicit bias. Panels (b)–(c) show the conserved leaves
316 that organize these paths for the endpoint geometries: for $p = 1$, the invariant is $s^2 - r^2$, whereas for
317 $p = \infty$, it is $\sqrt{s + r} + \sqrt{s - r}$ (see Appendix F for proofs). Panel (d) shows the impact of p in the
318 rank-two case over the (r_1, r_2) feature space.

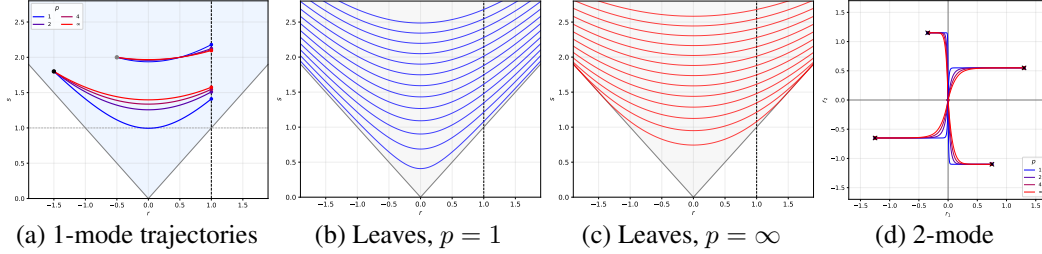


Figure 6: Closed-form Gaussian geometry. Panel (a): one-mode trajectories in (r, s) , for $p \in \{1, 2, 4, \infty\}$. Panels (b)–(c): conserved foliations for the two endpoint geometries. Panel (d): two-mode rank-two dynamics in the (r_1, r_2) plane; all trajectories start from the white-noise state $r_1 = r_2 = 0$, $(s_1, s_2) = (0.1, 10)$, and each group targets a different (β_1, β_2) .

319 **Geodesic convexity.** Geodesic convexity, i.e., the condition $\mathcal{F}(\mu_t) \leq (1 - t)\mathcal{F}(\mu_0) + t\mathcal{F}(\mu_1)$
320 along every constant-speed W_γ -geodesic $(\mu_t)_{t \in [0, 1]}$, is a central structural notion for metric gradient
321 flows [Ambrosio et al., 2008], as it underlies stability and global convergence results. The following
322 proposition summarizes several basic convexity examples; Appendix H contains the proofs and
323 a refined κ -geodesic convexity analysis. These results are, however, of limited direct use in the
324 machine-learning examples considered above, since the quadratic functionals arising in the MLP and
325 MMD settings are typically not geodesically convex. Note also that geodesic convexity of the entropy
326 is stated only along a selected geodesic: uniqueness of geodesics may fail, in particular for $p = \infty$,
327 which makes the “muon” setting is more delicate.

328 **Proposition 5.4** (Basic geodesic convexity examples). *Assume that γ is monotone. A linear functional*
329 *$\mathcal{F}(\mu) = \int h d\mu$, with $h : \mathbb{R}^d \rightarrow \mathbb{R}$, is W_γ -geodesically convex if and only if h is convex. A quadratic*
330 *interaction functional $\mathcal{F}(\mu) = \int h(x, y) d\mu(x)d\mu(y)$, with $h : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, is W_γ -geodesically*
331 *convex if h is convex. Finally, let $d\nu = Z^{-1}e^{-V}dx$, with $V \in C^2(\mathbb{R}^d)$ convex, and define the relative*
332 *entropy by $\mathcal{F}(\mu) = \int \log(d\mu/d\nu) d\mu$ if $\mu \ll \nu$, and $\mathcal{F}(\mu) = +\infty$ otherwise. If the representing*
333 *set $\mathcal{K}_\gamma$ contains at least one positive definite matrix, then for every $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$ there exists a*
334 *constant-speed W_γ -geodesic $(\mu_t)_{t \in [0, 1]}$ joining them along which $\mathcal{F}$ is convex.*

335 6 Conclusion

336 The main message of this paper is that matrix-normalized optimizers for mean-field neural models
337 are naturally encoded by Spectral Wasserstein distances. In this view, matrix normalization is not
338 merely a preconditioning device, but the local steepest-descent geometry generated by a spectral
339 transport cost. This clarifies the transport side, through coupling-based costs, geodesics, and Gaussian
340 reductions, and the optimization side, through a continuum interpretation of normalized matrix
341 updates. The resulting picture is that the choice of spectral gauge shapes which directions of motion
342 are emphasized or suppressed in the mean-field dynamics, thereby providing a geometric language
343 for comparing matrix-normalized optimizers. A major open question is the rigorous study of the
344 Spectral Wasserstein gradient-flow PDE: global-in-time existence, stability, and the avoidance of
345 local minima for shallow neural-network and MMD losses.

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426 A Matrix Recap and Explicit LMOs

427 This appendix collects the matrix facts used in Section 2. The canonical polar representing set
428 is $\mathcal{P}_\gamma := \{Q \in \mathbb{S}^d : \text{tr}(QS) \leq \gamma(S) \text{ for every } S \succeq 0\}$; finite-dimensional convex duality gives
429 $\gamma(S) = \max_{Q \in \mathcal{P}_\gamma} \text{tr}(QS)$ for every $S \succeq 0$.

430 We first shows that the use of positive gauges on the cone $\mathbb{S}_+^d$ is not essentially more restrictive than
431 working with norms on the full space of symmetric matrices: under monotonicity, every such gauge
432 admits a canonical norm extension.

433 **Proposition A.1** (Norm extension of a monotone positive gauge). *Let $\gamma : \mathbb{S}_+^d \rightarrow \mathbb{R}_+$ be a monotone*
434 *positive gauge, i.e., γ is convex, positively one-homogeneous, $\gamma(S) > 0$ for every nonzero $S \succeq 0$,*
435 *and $0 \preceq S \preceq S'$ implies $\gamma(S) \leq \gamma(S')$. Define, for $A \in \mathbb{S}^d$,*

$$\|A\|_\gamma := \inf \left\{ \gamma(P) + \gamma(N) : P, N \succeq 0, A = P - N \right\}.$$

436 *Then $\|\cdot\|_\gamma$ is a norm on $\mathbb{S}^d$, and its restriction to $\mathbb{S}_+^d$ coincides with γ .*

437 *Proof.* We first note that the admissible set is nonempty for every $A \in \mathbb{S}^d$: for instance, the spectral
 438 decomposition $A = A_+ - A_-$ gives $A_+, A_- \succeq 0$. Hence $\|A\|_\gamma$ is well-defined and finite.

439 Positive homogeneity follows directly from the definition. If $a \geq 0$, then

$$\|aA\|_\gamma = \inf_{A=P-N} \{\gamma(aP) + \gamma(aN)\} = a\|A\|_\gamma.$$

440 If $a < 0$, then $aA = (-a)N - (-a)P$ whenever $A = P - N$, so the same argument gives
 441 $\|aA\|_\gamma = |a|\|A\|_\gamma$. In particular, $\| -A \|_\gamma = \|A\|_\gamma$.

442 We next prove the triangle inequality. Let $A = P_1 - N_1$ and $B = P_2 - N_2$ with $P_i, N_i \succeq 0$. Then

$$A + B = (P_1 + P_2) - (N_1 + N_2),$$

443 and $P_1 + P_2, N_1 + N_2 \succeq 0$. By subadditivity of γ , which follows from convexity and one-homogeneity,

$$\gamma(P_1 + P_2) + \gamma(N_1 + N_2) \leq \gamma(P_1) + \gamma(P_2) + \gamma(N_1) + \gamma(N_2).$$

444 Taking the infimum over all decompositions of A and B yields $\|A + B\|_\gamma \leq \|A\|_\gamma + \|B\|_\gamma$.

445 It remains to prove nondegeneracy. Since γ is strictly positive on the compact section $\{S \succeq 0 : \text{tr}(S) = 1\}$, there exists $c > 0$ such that $\gamma(S) \geq c \text{tr}(S)$ for every $S \succeq 0$. Therefore, for every
 446 decomposition $A = P - N$,
 447

$$\gamma(P) + \gamma(N) \geq c \text{tr}(P + N).$$

448 Moreover, from $A = P - N$ we have $P + N \succeq A$ and $P + N \succeq -A$, which implies $\text{tr}(P + N) \geq$
 449 $\text{tr}|A|$. Hence

$$\gamma(P) + \gamma(N) \geq c \text{tr}|A|.$$

450 Taking the infimum gives $\|A\|_\gamma \geq c \text{tr}|A|$. Thus $\|A\|_\gamma = 0$ implies $A = 0$.

451 Finally, let $S \succeq 0$. The decomposition $S = S - 0$ gives $\|S\|_\gamma \leq \gamma(S)$. Conversely, if $S = P - N$
 452 with $P, N \succeq 0$, then $P = S + N \succeq S$, and by monotonicity, $\gamma(P) \geq \gamma(S)$. Hence

$$\gamma(P) + \gamma(N) \geq \gamma(P) \geq \gamma(S).$$

453 Taking the infimum over all such decompositions gives $\|S\|_\gamma \geq \gamma(S)$. Therefore $\|S\|_\gamma = \gamma(S)$ for
 454 every $S \succeq 0$. $\square$

455 The next proposition characterizes the monotone norms for which representing sets may be chosen
 456 inside the positive-semidefinite cone.

457 **Proposition A.2** (PSD representation and monotonicity). *The following are equivalent:*

- γ is monotone on $\mathbb{S}_+^d$, namely

$$0 \preceq S \preceq T \implies \gamma(S) \leq \gamma(T);$$

- there exists a convex compact representing set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ such that

$$\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS) \quad \text{for every } S \succeq 0.$$

Moreover, when these properties hold, the canonical choice

$$\mathcal{K}_\gamma = \mathcal{P}_\gamma \cap \mathbb{S}_+^d$$

458 is admissible.

459 *Proof.* If $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, then for $0 \preceq S \preceq T$ one has $\text{tr}(QS) \leq \text{tr}(QT)$ for every $Q \in \mathcal{K}_\gamma$, hence
 460 $\gamma(S) \leq \gamma(T)$ by taking suprema.

Conversely, assume γ is monotone. Define

$$\mathcal{K}_\gamma^+ := \{Q \in \mathbb{S}_+^d : \text{tr}(QS) \leq \gamma(S) \text{ for every } S \succeq 0\}.$$

It is immediate that $\mathcal{K}_\gamma^+$ is convex, closed, bounded, and contained in $\mathcal{P}_\gamma \cap \mathbb{S}_+^d$. We claim that it represents γ . Fix $S \succeq 0$. If $S \succ 0$, choose a subgradient $Q \in \partial\gamma(S)$ with respect to the ambient vector space of symmetric matrices. Since γ is convex and one-homogeneous, $\text{tr}(QS) = \gamma(S)$ and

$\text{tr}(QT) \leq \gamma(T)$ for every $T \succeq 0$. Moreover $Q \succeq 0$: indeed, for every $R \succeq 0$ and every sufficiently small $\varepsilon > 0$ such that $S - \varepsilon R \succeq 0$, the subgradient inequality gives

$$\gamma(S - \varepsilon R) \geq \gamma(S) - \varepsilon \text{tr}(QR),$$

while monotonicity gives $\gamma(S - \varepsilon R) \leq \gamma(S)$. Hence $\text{tr}(QR) \geq 0$ for all $R \succeq 0$, i.e. $Q \succeq 0$. Thus $Q \in \mathcal{K}_\gamma^+$ and $\gamma(S) \leq \sup_{P \in \mathcal{K}_\gamma^+} \text{tr}(PS)$. The reverse inequality follows from the definition of $\mathcal{K}_\gamma^+$. For singular $S \succeq 0$, apply the previous argument to $S + \varepsilon I$ and let $\varepsilon \downarrow 0$ using continuity of the norm. Therefore $\gamma(S) = \max_{Q \in \mathcal{K}_\gamma^+} \text{tr}(QS)$ for every $S \succeq 0$. Convexity and compactness follow from finite dimensionality. $\square$

Proposition A.3 (Norm property of $\mathcal{N}$). *Assume that γ is monotone and admits a positive semidefinite representing set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, so that $\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS)$ for every $S \succeq 0$. Define $\mathcal{N}(V) := \gamma(V^\top V)^{1/2}$. Then $\mathcal{N}$ is a seminorm. If, in addition, $\gamma(S) > 0$ for every nonzero $S \succeq 0$, then $\mathcal{N}$ is a norm.*

Proof. For every matrix V , one has

$$\mathcal{N}(V)^2 = \gamma(V^\top V) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QV^\top V) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(VQV^\top).$$

Since $Q \succeq 0$, we may write $\text{tr}(VQV^\top) = \|VQ^{1/2}\|_F^2$. Hence

$$\mathcal{N}(V) = \max_{Q \in \mathcal{K}_\gamma} \|VQ^{1/2}\|_F.$$

For each fixed $Q \in \mathcal{K}_\gamma$, the map $V \mapsto \|VQ^{1/2}\|_F$ is a seminorm. Therefore, for any matrices V, W and scalar a ,

$$\mathcal{N}(aV) = \max_{Q \in \mathcal{K}_\gamma} \|aVQ^{1/2}\|_F = |a|\mathcal{N}(V),$$

and

$$\mathcal{N}(V + W) = \max_{Q \in \mathcal{K}_\gamma} \|(V + W)Q^{1/2}\|_F \leq \max_{Q \in \mathcal{K}_\gamma} (\|VQ^{1/2}\|_F + \|WQ^{1/2}\|_F) \leq \mathcal{N}(V) + \mathcal{N}(W).$$

Thus $\mathcal{N}$ is a seminorm. If moreover $\gamma(S) > 0$ for every nonzero $S \succeq 0$, then $\mathcal{N}(V) = 0$ implies $\gamma(V^\top V) = 0$, hence $V^\top V = 0$, and therefore $V = 0$. Thus $\mathcal{N}$ is a norm. $\square$

Remark A.4 ($\mathcal{K}_\gamma$ for the Schatten p -norms). *For $\gamma = \gamma_p$, a convenient supporting set is obtained from the dual Schatten unit ball. Namely, if $p \in [1, \infty]$ and $q \in [1, \infty]$ is the conjugate exponent, $1/p + 1/q = 1$, one may take*

$$\mathcal{K}_{\gamma_p} = \mathcal{P}_{\gamma_p} \cap \mathbb{S}_+^d := \left\{ G \in \mathbb{S}_+^d : \|G\|_{S_q} \leq 1 \right\}.$$

One can, however, choose a smaller supporting set in the endpoint cases. For $p = 1$, since $\|A\|_{S_1} = \text{tr}(A)$ on $\mathbb{S}_+^d$, it is enough to take $\mathcal{K}_{\gamma_1} = \{I\}$. For $p = \infty$, one may take the set of density matrices,

$$\mathcal{K}_{\gamma_\infty} = \left\{ G \in \mathbb{S}_+^d : \text{tr}(G) = 1 \right\}.$$

The next proposition gives the explicit LMO associated with each Schatten geometry.

Proposition A.5 (Explicit Schatten selectors). *Let $\gamma = \gamma_p = \|\cdot\|_{S_p}$, let $r = 2p$, and let $q = \frac{r}{r-1} = \frac{2p}{2p-1}$ be the dual exponent. If*

$$G = U \text{diag}(\sigma_i) W^\top$$

is the singular value decomposition of a gradient matrix, then

$$J_p(G) = -\|G\|_{S_q}^{2-q} U \text{diag}(\sigma_i^{q-1}) W^\top.$$

In particular,

- $p = 1$: $J_1(G) = -G$, which recovers ordinary gradient flow;

485 • $p = 2$: $J_2(G) = -\|G\|_{S_{4/3}}^{2/3} U \operatorname{diag}(\sigma_i^{1/3}) W^\top$, which we call the *Frobenius-intermediate*
 486 *normalization*;

487 • $p = \infty$:

$$J_\infty(G) = -\|G\|_{S_1} U W^\top = -\operatorname{tr}((G^\top G)^{1/2}) G (G^\top G)^{\dagger/2}, \quad (3)$$

488 which is the *Muon normalization*.

489 *Proof.* The displayed formula is the negative gradient of $\frac{1}{2} \|G\|_{S_q}^2$ when $1 < q \leq 2$, while the
 490 endpoint $q = 1$ is interpreted through a subgradient LMO. $\square$

491 A.1 Proofs for measure LMOs

492 Proof of Theorem 2.5.

Proof. For $Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d$, define

$$\Phi_Q(v) := \int g \cdot v \, d\mu + \frac{1}{2} \int v^\top Q v \, d\mu.$$

Since $\gamma(R) = \max_{Q \in \mathcal{K}_\gamma} \operatorname{tr}(QR)$ for $R \succeq 0$, the Measure-LMO objective is

$$\inf_v \max_{Q \in \mathcal{K}_\gamma} \Phi_Q(v).$$

For fixed positive definite Q , the inner minimization is strictly convex and gives

$$v_Q(x) = -Q^{-1}g(x), \quad \inf_v \Phi_Q(v) = -\frac{1}{2} \operatorname{tr}(Q^{-1} \mathcal{S}_\mu(g)).$$

The saddle point condition is therefore obtained by choosing Q_μ^* to maximize this dual value, equivalently to minimize $\operatorname{tr}((Q_\mu^*)^{-1} \mathcal{S}_\mu(g))$. With $v_*(x) = -(Q_\mu^*)^{-1}g(x)$, convex duality gives

$$\int g \cdot v_* \, d\mu + \frac{1}{2} \gamma \left(\int v_* v_*^\top \, d\mu \right) = -\frac{1}{2} \operatorname{tr}((Q_\mu^*)^{-1} \mathcal{S}_\mu(g)),$$

493 which equals the dual optimum. Thus, v_* minimizes the Measure-LMO objective. The limiting
 494 statement follows by lower semicontinuity of the objective in $L^2(\mu)$ along any minimizing sequence
 495 whose associated velocities converge. $\square$

496 Proof of Proposition 2.6.

Proof. For empirical measures, the continuity equation reduces to the characteristic system for the atoms. The identity $\mathcal{N}_{\mu_X}(v)^2 = n^{-1} \mathbf{N}(V)^2$ and the normalized pairing $n^{-1} \operatorname{tr}(G^\top V)$ reduce the measure-level minimization in Definition 2.3 to

$$\operatorname{argmin}_V \left\{ \operatorname{tr}(G^\top V) + \frac{1}{2} \mathbf{N}(V)^2 \right\} = J_\gamma(G).$$

497 Stacking the particle velocities yields $\dot{X}_t = J_\gamma(G_t)$. If the matrix objective is $F_n(X) = n\mathcal{F}(\mu_X)$,
 498 then $\nabla F_n(X) = G_t$, giving the same ODE. If instead $F(X) = \mathcal{F}(\mu_X)$, then $\nabla F(X) = n^{-1}G_t$ and
 499 one obtains $\dot{X}_t = n^{-1}J_\gamma(G_t)$ by one-homogeneity of J_γ . This is exactly the same weak measure
 500 equation after rescaling time by n . $\square$

501 A.2 Comparison between W_2 and W_γ

502 Proof of Proposition 3.2.

503 *Proof.* Since γ is a norm on the finite-dimensional space $\mathbb{S}^d$, its restriction to the compact set $\{S \succeq$
 504 $0 : \operatorname{tr}(S) = 1\}$ is continuous and strictly positive. Hence there exist constants $0 < c_\gamma \leq C_\gamma < \infty$
 505 such that $c_\gamma \operatorname{tr}(S) \leq \gamma(S) \leq C_\gamma \operatorname{tr}(S)$ for every $S \succeq 0$.

506 Let $\pi \in \Pi(\mu, \nu)$ be any coupling, and define its displacement covariance by

$$\Sigma_\pi := \int_{\mathbb{R}^d \times \mathbb{R}^d} (x - y)(x - y)^\top d\pi(x, y).$$

507 Then $\Sigma_\pi \succeq 0$, and therefore

$$c_\gamma \operatorname{tr}(\Sigma_\pi) \leq \gamma(\Sigma_\pi) \leq C_\gamma \operatorname{tr}(\Sigma_\pi).$$

508 Moreover,

$$\operatorname{tr}(\Sigma_\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \operatorname{tr}((x - y)(x - y)^\top) d\pi(x, y) = \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\pi(x, y).$$

509 Thus, for every coupling $\pi \in \Pi(\mu, \nu)$,

$$c_\gamma \int |x - y|^2 d\pi(x, y) \leq \gamma(\Sigma_\pi) \leq C_\gamma \int |x - y|^2 d\pi(x, y).$$

510 Taking infima over $\pi \in \Pi(\mu, \nu)$ gives

$$c_\gamma W_2(\mu, \nu)^2 \leq W_\gamma(\mu, \nu)^2 \leq C_\gamma W_2(\mu, \nu)^2,$$

511 where we used the definitions of W_2 and W_γ . Taking square roots yields

$$\sqrt{c_\gamma} W_2(\mu, \nu) \leq W_\gamma(\mu, \nu) \leq \sqrt{C_\gamma} W_2(\mu, \nu).$$

512 □

513 **Schatten constants and sharpness.** For Schatten norms, $d^{1/p-1} W_2(\mu, \nu)^2 \leq W_{\gamma_p}(\mu, \nu)^2 \leq$
 514 $W_2(\mu, \nu)^2$, equivalently $c_{\gamma_p} = d^{1/p-1}$ and $C_{\gamma_p} = 1$. In particular, $(c_{\gamma_2}, C_{\gamma_2}) = (d^{-1/2}, 1)$ and
 515 $(c_{\gamma_\infty}, C_{\gamma_\infty}) = (d^{-1}, 1)$. These constants follow from $\|\lambda\|_{\ell_p} \leq \|\lambda\|_{\ell_1} \leq d^{1-1/p} \|\lambda\|_{\ell_p}$.

516 The lower Schatten bound is sharp. For $\gamma = \gamma_p$, equality in the lower bound is attained by isotropic
 517 Gaussian pairs $\mu = \mathcal{N}(0, \alpha I_d)$ and $\nu = \mathcal{N}(0, \beta I_d)$, $\alpha, \beta > 0$. By the Gaussian computation of
 518 Appendix D, the displacement covariance is $(\sqrt{\alpha} - \sqrt{\beta})^2 I_d$, hence $W_{\gamma_p}(\mu, \nu)^2 = d^{1/p-1} W_2(\mu, \nu)^2$.

519 Proof of Theorem 3.3.

520 *Proof.* Let $\pi \in \Pi(\mu, \nu)$ and set

$$\Sigma_\pi := \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)(y - x)^\top d\pi(x, y).$$

521 Since $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, the matrix Σ_π is well-defined and belongs to $\mathbb{S}_+^d$. By the support representation
 522 of γ ,

$$\gamma(\Sigma_\pi) = \max_{Q \in \mathcal{K}_\gamma} \operatorname{tr}(Q \Sigma_\pi).$$

523 Moreover, for every $Q \in \mathcal{K}_\gamma$,

$$\operatorname{tr}(Q \Sigma_\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)^\top Q (y - x) d\pi(x, y).$$

524 Therefore

$$W_\gamma(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \max_{Q \in \mathcal{K}_\gamma} \int (y - x)^\top Q (y - x) d\pi(x, y).$$

525 It remains to justify the exchange of $\inf_\pi$ and $\max_Q$. Define

$$F(\pi, Q) := \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)^\top Q (y - x) d\pi(x, y).$$

526 The coupling set $\Pi(\mu, \nu)$ is convex and weakly compact, since its marginals are fixed. The set $\mathcal{K}_\gamma$
 527 is convex and compact by assumption. For fixed π , the map $Q \mapsto F(\pi, Q)$ is affine, hence both
 528 concave and convex, and is continuous on $\mathcal{K}_\gamma$. For fixed $Q \succeq 0$, the cost $(x, y) \mapsto (y - x)^\top Q (y - x)$

is nonnegative and lower semicontinuous. Hence $\pi \mapsto F(\pi, Q)$ is lower semicontinuous for weak convergence of probability measures, by the Portmanteau theorem. It is also affine in π .

Thus the lower-semicontinuous form of Sion's minimax theorem Sion [1958] applies and gives

$$\inf_{\pi \in \Pi(\mu, \nu)} \max_{Q \in \mathcal{K}_\gamma} F(\pi, Q) = \max_{Q \in \mathcal{K}_\gamma} \inf_{\pi \in \Pi(\mu, \nu)} F(\pi, Q).$$

The only point requiring care is that the quadratic cost is not bounded on $\mathbb{R}^d \times \mathbb{R}^d$. This is precisely why one uses the lower-semicontinuous version of the minimax theorem rather than the bounded continuous one. Consequently,

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} \inf_{\pi \in \Pi(\mu, \nu)} \int (y - x)^\top Q(y - x) d\pi(x, y).$$

Finally, by definition of the quadratic optimal transport cost associated with Q ,

$$W^Q(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \int (y - x)^\top Q(y - x) d\pi(x, y).$$

Therefore

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu)^2,$$

which proves the claim. $\square$

A.3 Metric property from the robust-cost formulation

The max–min representation of Theorem 3.3 gives a direct proof of the triangle inequality when γ is monotone. This avoids using the Benamou–Brenier formulation for this specific purpose.

Proposition A.6 (Static proof of the metric property). *Assume that γ is monotone on $\mathbb{S}_+^d$. Then W_γ is a distance on $\mathcal{P}_2(\mathbb{R}^d)$ and induces the same topology as W_2 .*

Proof. By Proposition A.2, monotonicity allows us to choose a representing set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ such that

$$\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS), \quad S \succeq 0.$$

Hence Theorem 3.3 gives

$$W_\gamma(\mu, \nu) = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu).$$

For a fixed $Q \in \mathcal{K}_\gamma$, write

$$d_Q(x, y) := \|Q^{1/2}(x - y)\|_2.$$

Since $Q \succeq 0$, d_Q is a seminorm distance on $\mathbb{R}^d$, and therefore W^Q is the corresponding quadratic Wasserstein pseudodistance. In particular, for all $\mu_0, \mu_1, \mu_2 \in \mathcal{P}_2(\mathbb{R}^d)$,

$$W^Q(\mu_0, \mu_2) \leq W^Q(\mu_0, \mu_1) + W^Q(\mu_1, \mu_2).$$

Taking the maximum over $Q \in \mathcal{K}_\gamma$ yields

$$\begin{aligned} W_\gamma(\mu_0, \mu_2) &= \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu_0, \mu_2) \\ &\leq \max_{Q \in \mathcal{K}_\gamma} \{W^Q(\mu_0, \mu_1) + W^Q(\mu_1, \mu_2)\} \\ &\leq \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu_0, \mu_1) + \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu_1, \mu_2) \\ &= W_\gamma(\mu_0, \mu_1) + W_\gamma(\mu_1, \mu_2). \end{aligned}$$

Symmetry is immediate from the definition. Moreover $W_\gamma(\mu, \mu) = 0$, and separation follows from Proposition 3.2: if $W_\gamma(\mu, \nu) = 0$, then $W_2(\mu, \nu) = 0$, hence $\mu = \nu$. The same comparison bounds also show that W_γ and W_2 induce the same topology on $\mathcal{P}_2(\mathbb{R}^d)$. $\square$

Remark A.7 (Relation with subspace robust Wasserstein distances). *The proof is the same robust-cost argument used by Paty and Cuturi [2019] for Ky Fan gauges. The only point needed in their argument is that the maximizing cost matrices are positive semidefinite. By Proposition A.2, this positivity holds for any monotone gauge γ , not only for Ky Fan gauges. Thus the Paty–Cuturi triangle-inequality proof extends verbatim to the present monotone-gauge setting.*

557 A.4 Large-mean asymptotics for Schatten geometries

558 We now describe the behavior of W_γ when the displacement between the means is much larger than
 559 the centered fluctuations of the measures. This regime is useful to interpret the optimizer in the
 560 max-min formula: for Schatten geometries with $p > 1$, the optimal certificate becomes increasingly
 561 aligned with the direction of the mean displacement, and the residual transport problem is effectively
 562 measured only along this direction.

563 **Proposition A.8** (Large-mean asymptotics for Schatten geometries). *Let $\gamma = \gamma_p$ be the Schatten- p
 564 norm, with $p \in [1, \infty]$, and let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$. Set $m_\mu := \int x d\mu(x)$, $m_\nu := \int x d\nu(x)$, and
 565 $\Delta := m_\nu - m_\mu$. Assume $\Delta \neq 0$, and write $\Delta = Ru$, where $R = |\Delta|$ and $u = \Delta/|\Delta|$. Let
 566 $\bar{\mu} := (\text{Id} - m_\mu)_\# \mu$ and $\bar{\nu} := (\text{Id} - m_\nu)_\# \nu$. Then, as $R \rightarrow \infty$ with $\bar{\mu}, \bar{\nu}$ fixed, the following
 567 asymptotics hold. For $p = 1$,*

$$W_\gamma(\mu, \nu)^2 = R^2 + W_2(\bar{\mu}, \bar{\nu})^2.$$

568 For $1 < p \leq \infty$,

$$W_\gamma(\mu, \nu)^2 = R^2 + W^{uu^\top}(\bar{\mu}, \bar{\nu})^2 + o(1).$$

569 *Proof.* We first separate the mean displacement from the centered transport. Fix $Q \succeq 0$. If $\pi \in$
 570 $\Pi(\mu, \nu)$ and if $\bar{\pi}$ denotes the corresponding coupling of $\bar{\mu}$ and $\bar{\nu}$, then writing $x = m_\mu + \bar{x}$ and
 571 $y = m_\nu + \bar{y}$ gives

$$y - x = \Delta + (\bar{y} - \bar{x}).$$

572 Therefore

$$\begin{aligned} \int (y - x)^\top Q (y - x) d\pi(x, y) &= \Delta^\top Q \Delta + 2\Delta^\top Q \int (\bar{y} - \bar{x}) d\bar{\pi}(\bar{x}, \bar{y}) \\ &\quad + \int (\bar{y} - \bar{x})^\top Q (\bar{y} - \bar{x}) d\bar{\pi}(\bar{x}, \bar{y}). \end{aligned}$$

573 The middle term vanishes because $\bar{\mu}$ and $\bar{\nu}$ have zero mean. Taking the infimum over couplings yields
 574 the exact decomposition

$$W^Q(\mu, \nu)^2 = \Delta^\top Q \Delta + W^Q(\bar{\mu}, \bar{\nu})^2.$$

575 By Theorem 3.3,

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_{\gamma_p}} \{ \Delta^\top Q \Delta + W^Q(\bar{\mu}, \bar{\nu})^2 \}.$$

576 For $p = 1$, we use the reduced representing set $\mathcal{K}_{\gamma_1} = \{\text{Id}\}$. Then

$$W_\gamma(\mu, \nu)^2 = \Delta^\top \Delta + W^{\text{Id}}(\bar{\mu}, \bar{\nu})^2 = R^2 + W_2(\bar{\mu}, \bar{\nu})^2,$$

577 as claimed.

578 Assume now that $1 < p \leq \infty$, and let q be the conjugate exponent, with $q = 1$ when $p = \infty$. We use
 579 the positive dual Schatten representing set

$$\mathcal{K}_{\gamma_p} = \{Q \succeq 0 : \|Q\|_{S_q} \leq 1\}.$$

580 Since $\Delta = Ru$, the max-min formula becomes

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_{\gamma_p}} \{ R^2 u^\top Q u + W^Q(\bar{\mu}, \bar{\nu})^2 \}.$$

581 By Schatten duality, applied to the rank-one matrix $uu^\top$,

$$\max_{Q \in \mathcal{K}_{\gamma_p}} u^\top Q u = \max_{\substack{Q \succeq 0 \\ \|Q\|_{S_q} \leq 1}} \text{tr}(Q uu^\top) = \|uu^\top\|_{S_p} = 1.$$

582 For $1 < p < \infty$, the equality case in Hölder's inequality is unique and gives $Q = uu^\top$. For $p = \infty$,
 583 the constraint is $\text{tr}(Q) \leq 1$, and $u^\top Q u \leq \text{tr}(Q) \leq 1$; equality again forces $Q = uu^\top$. Thus the
 584 exposed face of $\mathcal{K}_{\gamma_p}$ in the direction $uu^\top$ is the singleton $\{uu^\top\}$ for every $1 < p \leq \infty$.

585 Let Q_R be an optimal certificate for the previous maximization problem. Since $\mathcal{K}_{\gamma_p}$ is compact, any
 586 sequence $R_n \rightarrow \infty$ admits a subsequence such that $Q_{R_n} \rightarrow Q_\infty \in \mathcal{K}_{\gamma_p}$. Dividing the optimality

relation by R_n^2 and passing to the limit shows that Q_∞ maximizes $Q \mapsto u^\top Q u$ over $\mathcal{K}_{\gamma_p}$. By the uniqueness of the exposed maximizer, $Q_\infty = uu^\top$. Hence every maximizing certificate satisfies

$$Q_R \longrightarrow uu^\top \quad \text{as } R \rightarrow \infty.$$

Since $Q \mapsto W^Q(\bar{\mu}, \bar{\nu})^2$ is continuous on $\mathcal{K}_{\gamma_p}$, we obtain

$$W^{Q_R}(\bar{\mu}, \bar{\nu})^2 \longrightarrow W^{uu^\top}(\bar{\mu}, \bar{\nu})^2.$$

Substituting this into the max-min formula gives

$$W_\gamma(\mu, \nu)^2 = R^2 + W^{uu^\top}(\bar{\mu}, \bar{\nu})^2 + o(1).$$

Taking square roots yields

$$W_\gamma(\mu, \nu) = R + \frac{1}{2R} W^{uu^\top}(\bar{\mu}, \bar{\nu})^2 + o(R^{-1}),$$

which completes the proof. $\square$

The proposition shows that all Schatten geometries have the same leading large-mean behavior, namely $W_\gamma(\mu, \nu) \sim |\Delta|$. The difference appears at the next order and in the optimizer selected by the max-min formula. For the reduced trace geometry $p = 1$, the certificate is fixed and isotropic, $Q = \text{Id}$, so the centered correction is the usual Wasserstein cost $W_2(\bar{\mu}, \bar{\nu})^2$. By contrast, for every $p > 1$, the optimal certificate aligns with the normalized mean displacement $u = \Delta/|\Delta|$, and the centered correction becomes $W^{uu^\top}(\bar{\mu}, \bar{\nu})^2$. Thus, in the large-mean regime, the coupling selected by the Schatten geometry is increasingly influenced by the one-dimensional projection along u : transverse fluctuations only affect the cost at lower order through the way they interact with the projected quadratic transport problem.

B Connection with Muon, Orthogonalization, and Momentum

This appendix complements Section 2 by linking the matrix ODE for $p = \infty$ to practical Muon variants used at scale, and by showing how Newton–Schulz orthogonalization and momentum fit in the same normalized-flow language.

Taking into account Newton–Schulz orthogonalization. Setting $p = \infty$ in Proposition A.5 yields the LMO (3), so the spectral matrix flow

$$\dot{X}_t = J_\infty(\nabla F(X_t))$$

is exactly the vanishing-step-size, vanishing-momentum, full-batch limit of idealized Muon in the normalization convention used throughout this paper.

An important practical feature of Muon is that one does not compute the exact polar factor $UW^\top$ by an SVD at every step. To make the method scalable and GPU-friendly, practical implementations replace $UW^\top = G(G^\top G)^{\dagger/2}$ by a few Newton–Schulz iterations [Jordan et al., 2024, Liu et al., 2025]. After rescaling G by a scalar $\alpha(G) > 0$ chosen so that the spectrum of $\alpha(G)^{-2}G^\top G$ lies in $[0, 1]$, the classical Newton–Schulz iteration reads

$$Z_{k+1} = \frac{1}{2}Z_k(3I - Z_k^\top Z_k), \quad Z_0 = \alpha(G)^{-1}G.$$

For the discussion below, let us freeze the scale and regard $\alpha > 0$ as fixed a priori. Then after one iteration one obtains

$$J_{\alpha,1}(G) := -Z_1 = -\frac{1}{2\alpha} G(3I - \alpha^{-2}G^\top G),$$

and after k iterations one obtains more generally a polynomial LMO of the form

$$J_{\alpha,k}(G) = -G p_{\alpha,k}(G^\top G),$$

where $p_{\alpha,k}$ is a matrix polynomial approximating $s \mapsto s^{-1/2}$ on the spectral interval of interest. In this fixed- α setting, the polynomial orthogonalization part does admit a scalar generating function: if

$$p_{\alpha,k}(s) = \sum_{\ell=0}^m a_\ell s^\ell,$$

then

$$\gamma_{\alpha,k}(S) := \sum_{\ell=0}^m \frac{a_\ell}{\ell+1} \text{tr}(S^{\ell+1})$$

satisfies

$$\nabla \gamma_{\alpha,k}(S) = p_{\alpha,k}(S) \quad \text{and} \quad J_{\alpha,k}(G) = -\nabla_G \left(\frac{1}{2} \gamma_{\alpha,k}(G^\top G) \right).$$

For one Newton–Schulz step this gives the explicit formula

$$\gamma_{\alpha,1}(S) = \frac{3}{2\alpha} \text{tr}(S) - \frac{1}{4\alpha^3} \text{tr}(S^2).$$

This function is not a norm, and it is not globally monotone on $\mathbb{S}_+^d$, because

$$\nabla \gamma_{\alpha,1}(S) = \frac{3}{2\alpha} I - \frac{1}{2\alpha^3} S$$

ceases to be positive semidefinite once S has eigenvalues larger than $3\alpha^2$. It is nevertheless monotone on the truncated spectral region

$$0 \preceq S \preceq 3\alpha^2 I,$$

608 and in particular on the standard Newton–Schulz regime $0 \preceq S \preceq \alpha^2 I$.

609 Therefore, if one replaces the exact Muon LMO J_∞ by the fixed- α Newton–Schulz LMO $J_{\alpha,k}$, the
610 resulting approximate Muon dynamics is again generated by a spectral functional, now $\gamma_{\alpha,k}$. The
611 main caveat is that in practical implementations the scaling parameter α is not frozen but chosen
612 adaptively from the current matrix in order to track the evolving spectrum during training. Extending
613 the present geometric interpretation to that state-dependent choice of α remains an open question.

614 **Taking into account momentum.** A natural momentum extension, in the spirit of Muon with ex-
615ponential moving average, is obtained by introducing an auxiliary matrix variable A_t and considering

$$\tau \dot{A}_t + A_t = \nabla F(X_t), \quad \dot{X}_t = J_\gamma(A_t), \quad (4)$$

616 where $\tau \geq 0$ is the memory time scale. The variable A_t plays the role of a continuous-time
617 exponentially averaged gradient. When $\tau = 0$, one recovers exactly the first-order spectral flow
618 studied above.

For smooth Schatten geometries ($1 < p < \infty$), this system is equivalent to a second-order inertial equation

$$\tau \frac{d}{dt} (\mathbb{I}_p(\dot{X}_t)) + \mathbb{I}_p(\dot{X}_t) + \nabla F(X_t) = 0,$$

where

$$\mathbb{I}_p(V) := \|V\|_{S_r}^{2-r} U \text{diag}(\sigma_i^{r-1}) W^\top, \quad r = 2p,$$

for $V = U \text{diag}(\sigma_i) W^\top$. In particular, for $p = 1$ one has $\mathbb{I}_1(V) = V$ and one recovers the heavy-ball equation

$$\tau \ddot{X}_t + \dot{X}_t + \nabla F(X_t) = 0.$$

At the endpoint $p = \infty$, (4) reads

$$\tau \dot{A}_t + A_t = \nabla F(X_t), \quad \dot{X}_t = -\|A_t\|_{S_1} U_t W_t^\top,$$

619 which is the exact-polar continuous-time Muon dynamics with momentum.

The corresponding mean-field description naturally lives on phase space. Let

$$\eta_t \in \mathcal{P}_2(\mathbb{R}^d \times \mathbb{R}^d)$$

be the law of pairs (x, a) , where x is the particle position and a is the exponentially averaged force. Its spatial marginal is

$$\mu_t = (\pi_x)_\# \eta_t.$$

For a functional $\mathcal{F}$ on measures, write its Wasserstein force as

$$g_\mu(x) = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(x).$$

Given a phase-space law η , define the momentum covariance

$$S(\eta) := \int_{\mathbb{R}^d \times \mathbb{R}^d} aa^\top d\eta(x, a).$$

In the monotone regime, and assuming for simplicity that an invertible inverse-trace minimizer can be selected, choose

$$Q_\eta^* \in \operatorname{argmin}_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \operatorname{tr}(Q^{-1}S(\eta)).$$

620 The phase-space characteristics are then

$$\dot{x}_t = -(Q_{\eta_t}^*)^{-1}a_t, \quad \tau \dot{a}_t = g_{\mu_t}(x_t) - a_t. \quad (5)$$

621 Consequently, the law η_t solves the Liouville-type equation

$$\partial_t \eta_t + \operatorname{div}_x(\eta_t u_t) + \frac{1}{\tau} \operatorname{div}_a(\eta_t(g_{\mu_t}(x) - a)) = 0, \quad (6)$$

where

$$u_t(x, a) = -(Q_{\eta_t}^*)^{-1}a, \quad \mu_t = (\pi_x)_\# \eta_t.$$

622 When $\tau = 0$, the averaged force instantaneously relaxes to $a = g_{\mu_t}(x)$, and the spatial marginal
623 formally recovers the first-order normalized PDE (2). For $\tau > 0$, the dynamics is no longer a metric
624 gradient flow for W_γ on the position marginal alone; it is a damped kinetic system in which the
625 spectral geometry enters through the constitutive relation between the phase-space force covariance
626 and the spatial velocity.

627 C Monge Maps and Conditional Brenier Theorem

628 This appendix collects the map-based statements associated with the static formulation of Section 3.

629 The next definition records the Monge restriction, which is useful for comparison but is not the correct
630 notion in general.

Definition C.1 (Monge restriction). *Define*

$$W_\gamma^M(\mu, \nu)^2 := \inf_{T \# \mu = \nu} \gamma \left(\int_{\mathbb{R}^d} (T(x) - x)(T(x) - x)^\top d\mu(x) \right),$$

631 with value $+\infty$ if no transport map exists.

632 The next proposition simply records that the map-based problem is a restriction of the coupling-based
633 one.

Proposition C.2 (Monge is a restriction). *For every $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$W_\gamma(\mu, \nu) \leq W_\gamma^M(\mu, \nu).$$

634 *Proof.* Every transport map T induces the coupling $(\operatorname{Id}, T)_\# \mu$, and the static cost evaluated on this
635 coupling is exactly the Monge cost associated with T . $\square$

636 The following remark shows on a two-Dirac example that the relaxation can be genuinely strict.

Remark C.3 (Strictness of the Monge restriction). *Consider*

$$\mu = \frac{1}{2}(\delta_{(-1,0)} + \delta_{(1,0)}), \quad \nu = \frac{1}{2}(\delta_{(0,-1)} + \delta_{(0,1)}).$$

Any transport map from μ to ν is a bijection between the two atoms. The two possible displacement covariance matrices are

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

so for the operator norm and Frobenius norm one gets

$$W_\gamma^M(\mu, \nu)^2 = 2.$$

By contrast, the split coupling assigning mass 1/4 to each source-target pair has displacement covariance equal to the identity matrix, hence

$$W_\gamma(\mu, \nu)^2 = 1 \quad \text{for } \gamma = \|\cdot\|_{S_\infty}, \quad W_\gamma(\mu, \nu)^2 = \sqrt{2} \quad \text{for } \gamma = \|\cdot\|_{S_2}.$$

637 *Therefore the inequality in Proposition C.2 is strict already for two-point measures.*

638 The next remark explains that, for a completely arbitrary norm on $\mathbb{S}_+^d$, the static cost need not yet be
639 a bona fide metric.

Remark C.4 (Triangle inequality may fail for non-monotone norms). *For Dirac masses, the unique coupling gives*

$$\mathbb{W}_\gamma(\delta_x, \delta_y)^2 = \gamma((y-x)(y-x)^\top).$$

Hence if $\mathbb{W}_\gamma$ were always a distance, then the pointwise cost

$$d_\gamma(x, y) := \sqrt{\gamma((y-x)(y-x)^\top)}$$

640 would in particular have to define a metric on $\mathbb{R}^d$.

This fails for general norms on $\mathbb{S}_+^d$. Fix $d \geq 2$ and a parameter $M > 2$, and define

$$\gamma_{\text{ns}}(S) := \text{tr}(S) + M \sum_{1 \leq i < j \leq d} |S_{ij}|, \quad S \succeq 0.$$

This is a norm on $\mathbb{S}_+^d$, but it is not spectral since it depends on the matrix entries and not only on the eigenvalues. For the displacement

$$\Delta = y - x = (\Delta_1, \dots, \Delta_d)$$

one has

$$d_{\gamma_{\text{ns}}}(x, y)^2 = \|\Delta\|_2^2 + M \sum_{1 \leq i < j \leq d} |\Delta_i \Delta_j|.$$

Taking

$$x_0 = 0, \quad x_1 = e_1, \quad x_2 = e_1 + e_2$$

gives

$$d_{\gamma_{\text{ns}}}(x_0, x_1) = 1, \quad d_{\gamma_{\text{ns}}}(x_1, x_2) = 1, \quad d_{\gamma_{\text{ns}}}(x_0, x_2) = \sqrt{2 + M} > 2,$$

641 so the triangle inequality fails.

By contrast, if γ is monotone on $\mathbb{S}_+^d$, then Proposition A.2 allows us to choose $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, and therefore

$$d_\gamma(x, y) = \sup_{Q \in \mathcal{K}_\gamma} \|Q^{1/2}(x-y)\|_2,$$

642 which is a supremum of seminorms and therefore a genuine distance on $\mathbb{R}^d$. For general measures,
643 however, this pointwise argument does not control the cross terms created by gluing couplings. The
644 full metric property of $\mathbb{W}_\gamma$ in the monotone case is therefore proved later, through the dynamic
645 formulation, in Corollary 4.3.

646 The following proposition explains when an optimal coupling is induced by a Monge map of Brenier
647 type.

Proposition C.5 (Conditional Brenier theorem). *Assume μ is absolutely continuous. If a maximizing matrix $Q_* \in \mathcal{K}_\gamma$ in Theorem 3.3 is positive definite, then every optimal coupling for $\mathbb{W}_\gamma(\mu, \nu)$ is induced by a map. More precisely, there exists a convex function u such that*

$$T_*(x) = Q_*^{-1/2} \nabla u(Q_*^{1/2} x)$$

648 is optimal.

649 For a general gauge γ , this hypothesis is genuinely restrictive: an active maximizer $Q_* \in \mathcal{K}_\gamma$ need
650 not be positive semidefinite, and may even be indefinite. If γ is monotone, however, Proposition A.2
651 allows one to choose the representing set $\mathcal{K}_\gamma$ inside $\mathbb{S}_+^d$; the active matrix is then automatically
652 positive semidefinite, and the remaining nondegeneracy assumption is simply that it be invertible. For
653 Schatten geometries with $1 \leq p < +\infty$, the condition $Q_* \in \partial\gamma(\Sigma_{\pi^*})$ suggests that, generically, Q_*
654 should be positive definite, so that a Brenier-type map is expected to exist. The endpoint $p = +\infty$
655 is sharply different: in that case $\gamma(\Sigma) = \lambda_{\max}(\Sigma)$ and Q_* is supported on the top eigenspace of
656 Σ_{π^*} . When the leading eigenvalue is simple, Q_* is rank one. Thus one should not generally expect
657 a Monge map for the associated degenerate quadratic cost; rather, it is natural to expect optimal
658 couplings to remain nonunique, and often diffuse, in directions orthogonal to the active eigenspace.

Proof. Fix such a positive definite maximizer Q_* . For any coupling π between μ and ν , let

$$\tilde{\pi} = (Q_*^{1/2}, Q_*^{1/2})_{\#} \pi,$$

and define

$$\tilde{\mu} = (Q_*^{1/2})_{\#} \mu, \quad \tilde{\nu} = (Q_*^{1/2})_{\#} \nu.$$

Then

$$\int (y - x)^\top Q_*(y - x) d\pi(x, y) = \int |\tilde{y} - \tilde{x}|^2 d\tilde{\pi}(\tilde{x}, \tilde{y}).$$

Because μ is absolutely continuous and $Q_*^{1/2}$ is invertible, $\tilde{\mu}$ is also absolutely continuous. Brenier's theorem for the quadratic cost therefore yields a convex potential u such that the unique optimal coupling between $\tilde{\mu}$ and $\tilde{\nu}$ is induced by the map

$$\tilde{T}(\tilde{x}) = \nabla u(\tilde{x}).$$

Pulling this map back to the original variables gives

$$T_*(x) = Q_*^{-1/2} \tilde{T}(Q_*^{1/2} x) = Q_*^{-1/2} \nabla u(Q_*^{1/2} x),$$

and the corresponding coupling is optimal for the inner problem associated with Q_* . Since Q_* maximizes Theorem 3.3, this coupling is also optimal for $W_\gamma(\mu, \nu)$. $\square$

D Gaussian Reductions

This appendix gathers the Gaussian reductions used in the static geometry and in the linear-network flow. Gaussian marginals compress the transport problem to a finite-dimensional optimization over admissible covariance blocks. The next proposition shows that, for Gaussian marginals, the infinite-dimensional transport problem collapses to an optimization over the cross-covariance matrix.

Proposition D.1 (Gaussian reduction). *Let $\mu = \mathcal{N}(m_0, \Sigma_0)$ and $\nu = \mathcal{N}(m_1, \Sigma_1)$. Then*

$$W_\gamma(\mu, \nu)^2 = \inf_K \gamma((m_1 - m_0)(m_1 - m_0)^\top + \Sigma_0 + \Sigma_1 - K - K^\top),$$

where the infimum runs over matrices K such that

$$\begin{pmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{pmatrix} \succeq 0.$$

In particular, for centered Gaussians the covariance cost

$$B_\gamma(\Sigma_0, \Sigma_1)^2 := \inf_{K: \begin{bmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{bmatrix} \succeq 0} \gamma(\Sigma_0 + \Sigma_1 - K - K^\top)$$

is well defined on the cone of covariance matrices. In the monotone regime of Section 4, it is the restriction of the metric W_γ to centered Gaussian laws and therefore defines a metric on covariance matrices.

Proof. Let (X, Y) be any coupling of the two Gaussian marginals. Every coupling has a cross-covariance $K = \mathbb{E}[(X - m_0)(Y - m_1)^\top]$ satisfying the block PSD constraint, the objective depends only on K , and conversely every feasible block covariance is realized by a jointly Gaussian coupling. Hence restricting to Gaussian couplings loses nothing. The covariance matrix of (X, Y) is

$$\begin{pmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{pmatrix}.$$

This block matrix must be positive semidefinite. Conversely, every such block positive semidefinite matrix defines a jointly Gaussian vector with marginals μ and ν .

For any admissible K , the displacement covariance equals

$$(m_1 - m_0)(m_1 - m_0)^\top + \Sigma_0 + \Sigma_1 - K - K^\top.$$

Therefore the generalized cost of a Gaussian coupling depends only on K . Optimizing over Gaussian couplings is exactly the same as optimizing over the block positive semidefinite constraint, which proves the formula. $\square$

674 The next corollary shows that commuting covariances lead to a closed form for the Schatten family,
 675 exactly as in the classical Bures case.

Corollary D.2 (Commuting covariances). *If Σ_0 and Σ_1 commute, then for Schatten norms γ_p one has*

$$B_{\gamma_p}(\Sigma_0, \Sigma_1)^2 = \gamma_p \left((\Sigma_0^{1/2} - \Sigma_1^{1/2})^2 \right).$$

Equivalently,

$$B_{\gamma_p}(\Sigma_0, \Sigma_1)^2 = \left(\sum_{i=1}^d \left| \sqrt{\lambda_i(\Sigma_0)} - \sqrt{\lambda_i(\Sigma_1)} \right|^{2p} \right)^{1/p}.$$

676 When $p = 1$, this is the usual Bures–Wasserstein formula.

677 *Proof.* If Σ_0 and Σ_1 commute, they are simultaneously diagonalizable. In that basis the block PSD
 678 constraint decouples coordinatewise, and the optimal choice is $K = \text{diag}(\sqrt{a_i}b_i)$. The resulting
 679 displacement covariance is $\text{diag}((\sqrt{a_i} - \sqrt{b_i})^2)$. $\square$

680 The following remark clarifies the scope of Corollary D.2.

681 **Remark D.3.** *The commuting formula is stated for Schatten norms because their value depends only*
 682 *on the eigenvalues of the displacement covariance. For more general norms on $\mathbb{S}_+^d$, even when Σ_0 and*
 683 *Σ_1 commute, one should not expect a closed form depending only on the eigenvalues $(\sqrt{a_i} - \sqrt{b_i})^2$,*
 684 *since the norm itself may retain basis-dependent information.*

685 E Dynamic Formulation Details

686 This appendix gathers the convex momentum formulation and the proofs deferred from Section 4.

687 E.1 Convex momentum reformulation

Passing to momenta linearizes the constraint. If $\mu = \rho\lambda$ and $m = w\lambda$ for some reference measure λ , define

$$\mathcal{A}_\gamma(\mu, m) := \sup_{Q \in \mathcal{K}_\gamma} \int_{\mathbb{R}^d} \frac{w(x)^\top Q w(x)}{\rho(x)} d\lambda(x),$$

688 with the usual perspective convention on $\{\rho = 0\}$.

689 The following proposition shows that the momentum formulation is convex and exactly matches the
 690 velocity action.

Proposition E.1 (Convex momentum action). *The action $\mathcal{A}_\gamma(\mu, m)$ is intrinsic, convex, and weak-* lower semicontinuous. If $m = \mu v$, then*

$$\mathcal{A}_\gamma(\mu, m) = \gamma \left(\int v(x) v(x)^\top d\mu(x) \right).$$

Hence

$$W_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 = \inf_{(\mu_t, m_t)} \int_0^1 \mathcal{A}_\gamma(\mu_t, m_t) dt$$

under the linear constraint

$$\partial_t \mu_t + \text{div}(m_t) = 0.$$

691 *Proof.* Let λ be any positive measure dominating both μ and m , and write $\mu = \rho\lambda$ and $m = w\lambda$,
 692 with $\rho \geq 0$ and $w : \mathbb{R}^d \rightarrow \mathbb{R}^d$. For $Q \succeq 0$, define the perspective quadratic integrand

$$\phi_Q(\rho, w) := \begin{cases} \frac{w^\top Q w}{\rho}, & \rho > 0, \\ 0, & \rho = 0 \text{ and } w = 0, \\ +\infty, & \rho = 0 \text{ and } w \neq 0. \end{cases}$$

Then ϕ_Q is convex and lower semicontinuous on $\mathbb{R}_+ \times \mathbb{R}^d$. Indeed, for $\rho > 0$ it is the perspective of the convex quadratic form $w \mapsto w^\top Q w$, and the above convention is its closed extension at $\rho = 0$.

The momentum action may be written as

$$\mathcal{A}_\gamma(\mu, m) = \sup_{Q \in \mathcal{K}_\gamma} \int \phi_Q(\rho(x), w(x)) d\lambda(x).$$

This formula is intrinsic, i.e., independent of the dominating measure λ . To see this, choose another dominating measure λ' and write $\mu = \rho' \lambda'$ and $m = w' \lambda'$. Passing to a common dominating measure, for instance $\eta = \lambda + \lambda'$, and using the one-homogeneity of ϕ_Q , namely $\phi_Q(a\rho, aw) = a\phi_Q(\rho, w)$ for $a \geq 0$, gives the same value of $\int \phi_Q d\lambda$ and $\int \phi_Q d\lambda'$. Thus $\mathcal{A}_\gamma(\mu, m)$ depends only on the pair (μ, m) .

We next prove the absolutely continuous representation. If $m = \mu v$, then one may take $\lambda = \mu$, so that $\rho = 1$ and $w = v$. Hence, for every $Q \in \mathcal{K}_\gamma$,

$$\int \phi_Q(1, v(x)) d\mu(x) = \int v(x)^\top Q v(x) d\mu(x) = \text{tr} \left(Q \int v(x) v(x)^\top d\mu(x) \right).$$

Taking the supremum over $Q \in \mathcal{K}_\gamma$ and using the support representation of γ yields

$$\mathcal{A}_\gamma(\mu, \mu v) = \sup_{Q \in \mathcal{K}_\gamma} \text{tr} \left(Q \int v v^\top d\mu \right) = \gamma \left(\int v v^\top d\mu \right).$$

Conversely, if $\mathcal{A}_\gamma(\mu, m) < +\infty$, then for every $Q \in \mathcal{K}_\gamma$ the integral $\int \phi_Q(\rho, w) d\lambda$ is finite. In particular, the singular case $\rho = 0$ and $w \neq 0$ cannot occur on a set of positive λ -measure. Thus m is absolutely continuous with respect to μ , and $m = \mu v$ with $v = w/\rho$ on $\{\rho > 0\}$.

Convexity follows from the same perspective formulation. For fixed $Q \succeq 0$, the map $(\mu, m) \mapsto \int \phi_Q(\rho, w) d\lambda$ is convex, because ϕ_Q is convex and one-homogeneous. Taking the supremum over $Q \in \mathcal{K}_\gamma$ preserves convexity, hence $(\mu, m) \mapsto \mathcal{A}_\gamma(\mu, m)$ is convex.

Finally, we prove weak-* lower semicontinuity. For each fixed $Q \succeq 0$, the functional

$$(\mu, m) \mapsto \int \phi_Q \left(\frac{d\mu}{d\lambda}, \frac{dm}{d\lambda} \right) d\lambda$$

is lower semicontinuous under weak-* convergence of measures, by the standard lower semicontinuity theorem for convex, lower semicontinuous, one-homogeneous integral functionals. Since $\mathcal{A}_\gamma$ is the supremum over $Q \in \mathcal{K}_\gamma$ of these lower semicontinuous functionals, $\mathcal{A}_\gamma$ is also weak-* lower semicontinuous.

The dynamic formula now follows by rewriting the velocity formulation in momentum variables. If (μ_t, v_t) satisfies $\partial_t \mu_t + \text{div}(\mu_t v_t) = 0$ and $m_t = \mu_t v_t$, then the constraint becomes the linear continuity equation $\partial_t \mu_t + \text{div}(m_t) = 0$, and the previous identity gives

$$\mathcal{N}_{\mu_t}(v_t)^2 = \gamma \left(\int v_t v_t^\top d\mu_t \right) = \mathcal{A}_\gamma(\mu_t, m_t).$$

Therefore the dynamic action $\int_0^1 \mathcal{N}_{\mu_t}(v_t)^2 dt$ is exactly $\int_0^1 \mathcal{A}_\gamma(\mu_t, m_t) dt$ in momentum variables. Taking the infimum over all admissible curves gives

$$\mathcal{W}_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 = \inf_{(\mu_t, m_t)} \int_0^1 \mathcal{A}_\gamma(\mu_t, m_t) dt,$$

under the linear constraint $\partial_t \mu_t + \text{div}(m_t) = 0$. □

E.2 Proof of Theorem 4.2 (static/dynamic equivalence)

Proof. Let $\pi \in \Pi(\mu_0, \mu_1)$ and consider the displacement interpolation

$$\mu_t = ((1-t)x + ty)_\# \pi.$$

The velocity along each segment is $y - x$, so

$$\int v_t v_t^\top d\mu_t = \int (y - x)(y - x)^\top d\pi$$

for all t . Therefore

$$W_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 \leq W_\gamma(\mu_0, \mu_1)^2.$$

Conversely, take any admissible dynamic plan (μ_t, v_t) . By the superposition principle, there exists a probability measure η on absolutely continuous paths γ such that

$$\mu_t = (e_t)_\# \eta, \quad \dot{\gamma}_t = v_t(\gamma_t)$$

for η -a.e. path and a.e. t . Let $\pi = (e_0, e_1)_\# \eta$. For every $Q \in \mathcal{K}_\gamma$,

$$\int_0^1 \int v_t^\top Q v_t d\mu_t dt = \int \int_0^1 \dot{\gamma}_t^\top Q \dot{\gamma}_t dt d\eta(\gamma).$$

This is the key point where monotonicity is used: by Proposition A.2 we have chosen $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, so every test matrix satisfies $Q \succeq 0$. Without this reduction one would have to work with possibly indefinite matrices, and the quadratic Jensen inequality below would no longer apply. Applying Jensen to the scalar function $t \mapsto Q^{1/2} \dot{\gamma}_t$ gives

$$\int_0^1 \dot{\gamma}_t^\top Q \dot{\gamma}_t dt \geq (\gamma_1 - \gamma_0)^\top Q (\gamma_1 - \gamma_0).$$

Hence

$$\int_0^1 \gamma \left(\int v_t v_t^\top d\mu_t \right) dt \geq \int (y - x)^\top Q (y - x) d\pi(x, y)$$

for every $Q \in \mathcal{K}_\gamma$. Taking the supremum over Q yields

$$\int_0^1 \gamma \left(\int v_t v_t^\top d\mu_t \right) dt \geq \gamma \left(\int (y - x)(y - x)^\top d\pi \right).$$

722 Finally, take the infimum over admissible dynamic plans. □

723 E.3 Proof of Corollary 4.4 (W_γ geodesics)

Proof. The upper bound follows by restricting the same displacement interpolation to the time interval $[s, t]$ and reparameterizing it to $[0, 1]$. Indeed, the rescaled velocity is $(t - s)(y - x)$, so Theorem 4.2 gives

$$W_\gamma(\mu_s, \mu_t)^2 \leq (t - s)^2 \gamma \left(\int (y - x)(y - x)^\top d\pi_*(x, y) \right) = (t - s)^2 W_\gamma(\mu_0, \mu_1)^2.$$

Applying the same argument to the segments $[0, s]$ and $[t, 1]$ yields

$$W_\gamma(\mu_0, \mu_s) \leq s W_\gamma(\mu_0, \mu_1), \quad W_\gamma(\mu_t, \mu_1) \leq (1 - t) W_\gamma(\mu_0, \mu_1).$$

By the triangle inequality from Corollary 4.3,

$$W_\gamma(\mu_0, \mu_1) \leq W_\gamma(\mu_0, \mu_s) + W_\gamma(\mu_s, \mu_t) + W_\gamma(\mu_t, \mu_1) \leq W_\gamma(\mu_s, \mu_t) + (1 - (t - s)) W_\gamma(\mu_0, \mu_1).$$

Rearranging gives

$$W_\gamma(\mu_s, \mu_t) \geq (t - s) W_\gamma(\mu_0, \mu_1),$$

724 hence equality holds. □

725 E.4 Proof of Proposition 5.1 (W_γ gradient flow)

726 *Proof.* We argue at the formal level, assuming enough regularity to justify the differentiations,
 727 integrations by parts, and infinitesimal identifications below. Let $t \mapsto \mu_t$ be represented by a
 728 continuity equation $\partial_t \mu_t + \text{div}(\mu_t v_t) = 0$. By the dynamic formulation of Theorem 4.2, the metric
 729 speed associated with W_γ is the spectral tangent norm, hence

$$|\mu'|_\gamma(t) = \mathcal{N}_{\mu_t}(v_t).$$

730 In particular, the tangent space at μ is equipped with the norm $v \mapsto \mathcal{N}_\mu(v)$.

731 We next identify the metric slope. For a tangent perturbation generated by w , namely $\partial_s \mu_s +$
 732 $\operatorname{div}(\mu_s w) = 0$ at $s = 0$ with $\mu_0 = \mu$, the first variation formula gives

$$\frac{d}{ds} \mathcal{F}(\mu_s) \Big|_{s=0} = \int g_\mu(x) \cdot w(x) d\mu(x), \quad g_\mu = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}.$$

733 Therefore the descending local slope is the dual norm of the linear functional $w \mapsto - \int g_\mu \cdot w d\mu$
 734 with respect to $\mathcal{N}_\mu$, namely

$$|\partial \mathcal{F}|_\gamma(\mu) = \sup_{\mathcal{N}_\mu(w) \leq 1} \left\{ - \int g_\mu(x) \cdot w(x) d\mu(x) \right\}.$$

735 Equivalently, by Fenchel duality,

$$\frac{1}{2} |\partial \mathcal{F}|_\gamma(\mu)^2 = \sup_w \left\{ - \int g_\mu(x) \cdot w(x) d\mu(x) - \frac{1}{2} \mathcal{N}_\mu(w)^2 \right\}.$$

736 The maximizer in this dual formulation is the same as the minimizer

$$\mathcal{J}_\gamma^\mu(g_\mu) \in \operatorname{argmin}_w \left\{ \int g_\mu(x) \cdot w(x) d\mu(x) + \frac{1}{2} \mathcal{N}_\mu(w)^2 \right\},$$

737 which is precisely the Measure-LMO of Definition 2.3.

738 Now take $v_t = \mathcal{J}_\gamma^{\mu_t}(g_{\mu_t})$. The optimality relation gives

$$\int g_{\mu_t} \cdot v_t d\mu_t + \frac{1}{2} \mathcal{N}_{\mu_t}(v_t)^2 = -\frac{1}{2} |\partial \mathcal{F}|_\gamma(\mu_t)^2.$$

739 Moreover, since the functional being minimized is $w \mapsto \int g_{\mu_t} \cdot w d\mu_t + \frac{1}{2} \mathcal{N}_{\mu_t}(w)^2$, the minimizer
 740 realizes the Riesz-type identity $\mathcal{N}_{\mu_t}(v_t) = |\partial \mathcal{F}|_\gamma(\mu_t)$. Hence

$$\int g_{\mu_t} \cdot v_t d\mu_t = -\frac{1}{2} \mathcal{N}_{\mu_t}(v_t)^2 - \frac{1}{2} |\partial \mathcal{F}|_\gamma(\mu_t)^2.$$

741 Using the chain rule along the continuity equation,

$$\frac{d}{dt} \mathcal{F}(\mu_t) = \int g_{\mu_t}(x) \cdot v_t(x) d\mu_t(x),$$

742 and the metric-speed identity $|\mu'|_\gamma(t) = \mathcal{N}_{\mu_t}(v_t)$, we obtain

$$\frac{d}{dt} \mathcal{F}(\mu_t) = -\frac{1}{2} |\mu'|_\gamma(t)^2 - \frac{1}{2} |\partial \mathcal{F}|_\gamma(\mu_t)^2.$$

743 This is exactly the energy dissipation identity for a curve of maximal slope. Finally, substituting
 744 $v_t = \mathcal{J}_\gamma^{\mu_t}(g_{\mu_t})$ into the continuity equation gives (2), completing the formal identification. $\square$

745 **F Gaussian-Preserving Flows and Linear Networks**

746 This section isolates a tractable regime in which the spectral flow reduces from an infinite-dimensional
 747 PDE to finite-dimensional covariance dynamics. The general Gaussian-preservation principle comes
 748 first, and the rest of the section specializes it to centered Gaussian training of linear two-layer
 749 networks.

750 **F.1 Gaussian-Preserving Gradient Flows**

751 This subsection isolates a simple class of spectral gradient flows that can still be analyzed rather
 752 explicitly. In general, understanding the long-time behavior of the flow is hard, but when Gaussian
 753 measures are preserved the infinite-dimensional evolution reduces to a finite-dimensional ODE on
 754 means and covariances.

755 The next corollary explains why Gaussian preservation transfers from the classical W_2 flow to the
 756 spectral flow whenever the inverse-trace minimizer remains invertible on the Gaussian class.

Corollary F.1 (From W_2 Gaussian preservation to spectral Gaussian preservation). *Assume that for every Gaussian state μ the classical Wasserstein gradient*

$$g_\mu(x) = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(x)$$

is affine in x , and that there exists an invertible inverse-trace minimizer

$$Q_\mu^* \in \operatorname{argmin}_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \operatorname{tr}(Q^{-1} \mathcal{S}_\mu(g_\mu)).$$

Then the spectral flow

$$\partial_t \mu_t + \operatorname{div}(\mu_t \mathcal{J}_\gamma^{\mu_t}(g_{\mu_t})) = 0$$

757 *preserves Gaussian measures.*

Proof. Fix a Gaussian state μ and write

$$g_\mu(x) = b_\mu + B_\mu x$$

for the affine W_2 gradient provided by the assumption. By Theorem 2.5, any invertible inverse-trace minimizer

$$Q_\mu^* \in \operatorname{argmin}_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \operatorname{tr}(Q^{-1} \mathcal{S}_\mu(g_\mu))$$

defines a valid spectral selector through

$$\mathcal{J}_\gamma^\mu(g_\mu)(x) = -(Q_\mu^*)^{-1}(b_\mu + B_\mu x).$$

This is again an affine vector field in x . Therefore, the spectral continuity equation is of the form

$$v_t(x) = a_t + A_t x$$

with coefficients depending on the current Gaussian state. A continuity equation with affine drift preserves the Gaussian class, and the associated mean m_t and covariance Σ_t solve the closed ODE system

$$\dot{m}_t = a_t + A_t m_t, \quad \dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top.$$

758 Therefore, Gaussian initial data remain Gaussian along the spectral flow. $\square$

759 Two important examples fit into this mechanism. First, for the relative entropy $\operatorname{Ent}_\nu(\mu)$ with Gaussian
760 target ν , the classical W_2 gradient flow is the Fokker–Planck equation, and on the Gaussian class its
761 Wasserstein gradient is affine in x . Second, whenever $\mathcal{F}(\mu)$ depends only on the mean and covariance
762 of μ , its first variation is a quadratic polynomial and its W_2 gradient is affine, so the same transfer
763 principle applies. This is in particular the case for training a linear two-layer network, obtained by
764 taking $\sigma = \operatorname{Id}$ in the MLP model above together with a quadratic loss $R(H)$: the resulting objective
765 depends quadratically on the first two moments of μ .

766 F.2 Centered Gaussian Linear Two-Layer Model

This subsection develops a concrete Gaussian-preserving example where the predictor is a linear map and the spectral flow closes on covariance matrices. Consider particles

$$x = (u, v) \in \mathbb{R}^d \times \mathbb{R}^d,$$

and the linear two-layer feature map

$$\phi(z, x) = (u^\top z)v, \quad z \in \mathbb{R}^d.$$

For a probability measure μ on (u, v) , the predictor is

$$H_\mu(z) = \int (u^\top z)v d\mu(u, v) = M_\mu z, \quad M_\mu := \int v u^\top d\mu(u, v).$$

With a linear teacher $y^*(z) = A_* z$ and data law ρ , the population least-squares loss is

$$\mathcal{F}(\mu) = \frac{1}{2} \int |H_\mu(z) - A_* z|^2 d\rho(z).$$

Writing

$$C_\rho := \int z z^\top d\rho(z), \quad B_\rho := \int y^*(z) z^\top d\rho(z) = A_\star C_\rho,$$

one obtains

$$\mathcal{F}(\mu) = \frac{1}{2} \text{tr}(M_\mu C_\rho M_\mu^\top) - \text{tr}(B_\rho M_\mu^\top) + \text{const.}$$

For centered Gaussian states

$$\mu = \mathcal{N}(0, \Sigma), \quad \Sigma = \begin{pmatrix} \Sigma_U & \Sigma_{UV} \\ \Sigma_{VU} & \Sigma_V \end{pmatrix},$$

767 the predictor is simply $M_\mu = \Sigma_{VU}$. Thus the cross-covariance block carries the learned linear map,
768 while the diagonal covariance blocks influence the normalized metric direction.

769 The next proposition gives the closed covariance equation for Schatten spectral flows.

Proposition F.2 (Covariance ODE for centered Gaussian states). *Let $\gamma = \gamma_p$ be the Schatten- p norm and let*

$$q = \frac{2p}{2p-1}$$

be the dual exponent of $2p$. Assume that $\mu_t = \mathcal{N}(0, \Sigma_t)$ is a centered Gaussian solution of the corresponding spectral flow and that Σ_t remains positive definite. Then

$$\dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top, \quad A_t = B_t \Sigma_t^{-1/2},$$

where

$$B_t = -\|K_t\|_{S_q}^{2-q} U_t \text{diag}(\sigma_i(t)^{q-1}) W_t^\top, \quad K_t = L_t \Sigma_t^{1/2},$$

$K_t = U_t \text{diag}(\sigma_i(t)) W_t^\top$ is a singular value decomposition, and

$$L_t = \begin{pmatrix} 0 & E_t^\top \\ E_t & 0 \end{pmatrix}, \quad E_t = \Sigma_{VU,t} C_\rho - B_\rho.$$

Equivalently,

$$\dot{\Sigma}_t = B_t \Sigma_t^{1/2} + \Sigma_t^{1/2} B_t^\top.$$

Proof. For centered Gaussian $\mu = \mathcal{N}(0, \Sigma)$, the loss depends on μ only through $M_\mu = \Sigma_{VU}$, so differentiating with respect to this block gives the matrix error $E = \Sigma_{VU} C_\rho - B_\rho$. Therefore

$$g_\mu(u, v) = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(u, v) = \begin{pmatrix} E^\top v \\ Eu \end{pmatrix} = Lx.$$

The Wasserstein gradient is affine, hence the Gaussian class is preserved by the mechanism of Corollary F.1. For an affine velocity $v(x) = Ax$, the selected direction solves

$$\min_A \left\{ \text{tr}(L \Sigma A^\top) + \frac{1}{2} \|A \Sigma^{1/2}\|_{S_{2p}}^2 \right\}.$$

Setting $B = A \Sigma^{1/2}$ and $K = L \Sigma^{1/2}$ reduces this to the finite-dimensional Schatten duality-map problem

$$\min_B \left\{ \text{tr}(KB^\top) + \frac{1}{2} \|B\|_{S_{2p}}^2 \right\},$$

770 whose minimizer is the displayed formula for B_t . Finally, if X_t follows $\dot{X}_t = A_t X_t$ and $X_t \sim$
771 $\mathcal{N}(0, \Sigma_t)$, then $\dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top$. $\square$

772 F.3 Modal Reduction Under Simultaneous Diagonalization

This subsection shows that, under a compatible data basis and balanced initialization, the covariance ODE decomposes into interpretable feature modes. Assume that after orthogonal changes of coordinates one has

$$C_\rho = \text{Id}, \quad B_\rho = \text{diag}(\beta_1, \dots, \beta_d),$$

with only finitely many nonzero coefficients, and use the balanced white-noise initialization

$$\Sigma_{U,0} = \alpha_0 \text{Id}, \quad \Sigma_{V,0} = \alpha_0 \text{Id}, \quad \Sigma_{UV,0} = 0, \quad \alpha_0 > 0.$$

Then the covariance remains diagonal mode by mode. We write

$$r_i(t) = (\Sigma_{VU,t})_{ii}, \quad s_i(t) = (\Sigma_{U,t})_{ii} = (\Sigma_{V,t})_{ii},$$

773 so that r_i is the predictor coefficient and s_i measures the corresponding layer magnitude.

Proposition F.3 (Modal ODE). *Under the assumptions above, set*

$$a_i = s_i + r_i, \quad b_i = s_i - r_i, \quad \mathcal{N}_t^q = \sum_{j=1}^d |r_j - \beta_j|^q \left(a_j^{q/2} + b_j^{q/2} \right).$$

Then each mode solves

$$\begin{aligned} \dot{r}_i &= -\mathcal{N}_t^{2-q} (r_i - \beta_i) |r_i - \beta_i|^{q-2} \left(a_i^{q/2} + b_i^{q/2} \right), \\ \dot{s}_i &= -\mathcal{N}_t^{2-q} (r_i - \beta_i) |r_i - \beta_i|^{q-2} \left(a_i^{q/2} - b_i^{q/2} \right). \end{aligned}$$

Equivalently,

$$\begin{aligned} \dot{a}_i &= -2\mathcal{N}_t^{2-q} (r_i - \beta_i) |r_i - \beta_i|^{q-2} a_i^{q/2}, \\ \dot{b}_i &= 2\mathcal{N}_t^{2-q} (r_i - \beta_i) |r_i - \beta_i|^{q-2} b_i^{q/2}. \end{aligned}$$

Proof. When Σ_{VU} is diagonal, the error $E_t = \Sigma_{VU,t} - B_\rho$ is diagonal, hence the affine gradient matrix

$$L_t = \begin{pmatrix} 0 & E_t \\ E_t & 0 \end{pmatrix}$$

preserves each coordinate plane. The balanced diagonal covariance structure is therefore invariant. In the i -th plane, the covariance block is

$$\Sigma_t^{(i)} = \begin{pmatrix} s_i & r_i \\ r_i & s_i \end{pmatrix},$$

774 with eigenvalues $a_i = s_i + r_i$ and $b_i = s_i - r_i$, while the corresponding block of L_t has eigenvalues
 775 $\pm(r_i - \beta_i)$. The singular values of $K_t = L_t \Sigma_t^{1/2}$ are therefore $|r_i - \beta_i| a_i^{1/2}$ and $|r_i - \beta_i| b_i^{1/2}$, with
 776 no additional factor of two. Applying the Schatten LMO of Proposition F.2 blockwise and collecting
 777 the common normalization gives the stated equations. $\square$

778 F.4 Rank-One and Rank-Two Targets

This subsection records the two simplest modal systems, which are useful for interpreting how the spectral normalization changes learning profiles. If only one target coefficient is nonzero,

$$B_\rho = \text{diag}(\beta, 0, \dots, 0), \quad \beta \neq 0,$$

then all inactive modes remain frozen and the dynamics reduces to the single pair (r, s) governed by Proposition F.3. The variables $a = s + r$ and $b = s - r$ remain positive and encode the two layer variances in the active symmetric and antisymmetric directions. This one-mode system is the basic phase portrait behind the Gaussian experiments: the predictor coefficient r_t moves toward β , while the terminal value of s_t records the implicit covariance bias selected by p . The admissible state set is the cone

$$\{(r, s) \in \mathbb{R}^2 : s > |r|\},$$

779 equivalently $a > 0$ and $b > 0$. In panel (a) of Figure 6 we display trajectories in this cone from two
 780 initial conditions. The next proposition gives the conserved leaves for the two endpoint geometries;
 781 these level sets organize the trajectories observed in panel (a).

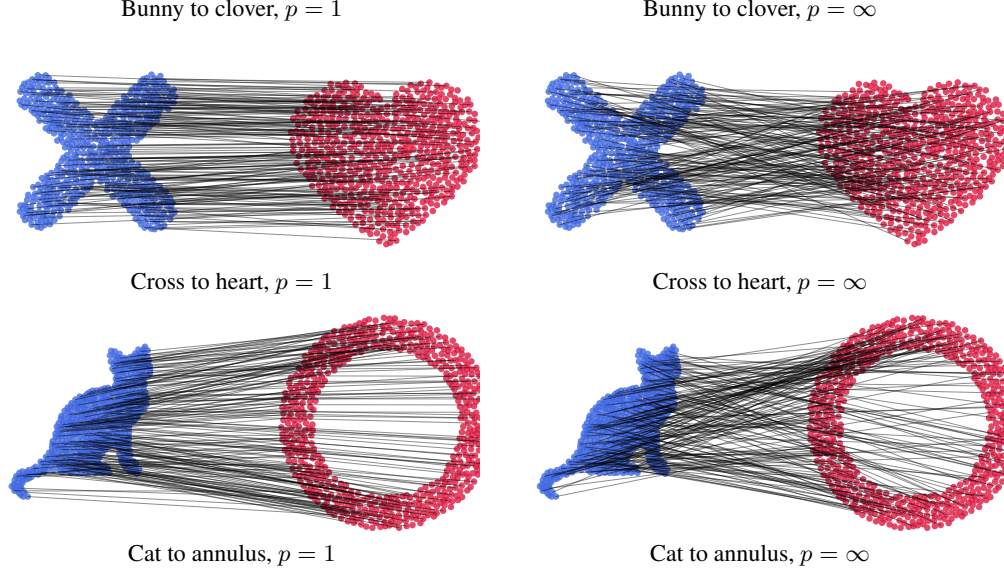


Figure 7: Additional static Spectral Wasserstein couplings. Only a subsample of matching segments is displayed.

Proposition F.4 (Rank-one endpoint invariants). *In the rank-one system, the trace geometry $p = 1$ preserves $s^2 - r^2$, while the operator geometry $p = \infty$ preserves $\sqrt{s} + r + \sqrt{s} - r$.*

Proof. Write $a = s + r$ and $b = s - r$. The modal equations of Proposition F.3 give $\dot{a} = -2\Lambda a^{q/2}$ and $\dot{b} = 2\Lambda b^{q/2}$ for a scalar factor Λ depending on the current state. For $p = 1$, one has $q = 2$, hence $\frac{d}{dt}(ab) = \dot{a}b + a\dot{b} = 0$, and $ab = s^2 - r^2$. For $p = \infty$, one has $q = 1$, hence $\frac{d}{dt}(\sqrt{a} + \sqrt{b}) = \dot{a}/(2\sqrt{a}) + \dot{b}/(2\sqrt{b}) = 0$, which gives the stated invariant. $\square$

Figure 6 in the main text illustrates these rank-one and rank-two reductions.

For a rank-two target

$$B_p = \text{diag}(\beta_1, \beta_2, 0, \dots, 0),$$

the first two modes solve the same equations but are coupled through

$$\mathcal{N}_t^q = \sum_{j=1}^2 |r_j - \beta_j|^q \left(a_j^{q/2} + b_j^{q/2} \right).$$

Thus the trace geometry evolves the two learned directions independently, whereas every $p > 1$ geometry shares a global normalization across them. This is the simplest Gaussian setting where the matrix nature of the spectral flow is visible at the level of the predictor rather than only through a time rescaling of a single scalar mode. In Figure 6(b), this coupling appears as distinct planar paths toward different targets (β_1, β_2) under the same initialization.

G Additional Coupling and Interpolation Examples

This section complements Figures 1 and 2 with the same static and displacement-interpolation experiments on additional pairs of planar shapes. Each example uses an image-defined source shape and a translated image-defined target shape.

Figure 7 shows additional static couplings for three image-defined source and target pairs, complementing the annulus-to-disks example in the main text.

Figure 8 displays the corresponding displacement-interpolation density snapshots, using the same optimal couplings as in Figure 7.

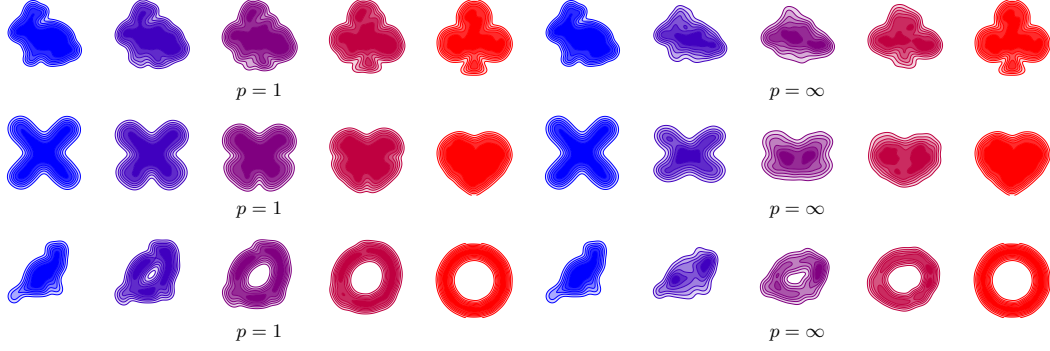


Figure 8: Additional displacement-interpolation density snapshots for the same shape pairs as Figure 7.

H Geodesic Convexity

This section studies one of the basic structural notions behind metric gradient flows, namely geodesic convexity, for the Spectral Wasserstein geometries introduced above. Geodesic convexity is desirable because it supports stability and global convergence statements [Ambrosio et al., 2008], but for many machine-learning objectives it is arguably too strong, so here it is used as a structural benchmark rather than as a blanket modeling assumption.

H.1 Definition and General Setup

This subsection recalls the relevant notion of convexity along Spectral Wasserstein geodesics and explains why it matters for the variational analysis of the flow. When one studies gradient flows in metric spaces, geodesic convexity plays the same role as ordinary convexity in Euclidean optimization: it governs uniqueness, stability, and the variational structure of the dynamics. Since Corollary 4.4 shows that W_γ -geodesics are displacement interpolations associated with optimal couplings, the corresponding convexity inequalities can be tested directly along Euclidean segments.

The next definition fixes the notion used in the remainder of this section.

Definition H.1 (Geodesic convexity). *A functional $F : \mathcal{P}_2(\mathbb{R}^d) \rightarrow (-\infty, +\infty]$ is called κ -geodesically convex for W_γ if along every constant-speed W_γ -geodesic $(\mu_t)_{t \in [0,1]}$ one has*

$$F(\mu_t) \leq (1-t)F(\mu_0) + tF(\mu_1) - \frac{\kappa}{2}t(1-t)W_\gamma(\mu_0, \mu_1)^2.$$

H.2 Linear Functionals

This subsection shows that for linear functionals the qualitative and quantitative convexity criteria reduce to explicit pointwise conditions on the integrand. For a Borel function $h : \mathbb{R}^d \rightarrow \mathbb{R}$, define the linear functional

$$F_h(\mu) := \int_{\mathbb{R}^d} h(x) d\mu(x).$$

Because W_γ -geodesics are still given by affine interpolation in space, the convexity question for F_h is especially transparent.

The next theorem gives the exact criterion.

Theorem H.2 (Linear functionals). *Assume γ is monotone and let $\mathcal{K}_\gamma$ be a convex compact PSD representing set for γ as in Proposition A.2.*

1. *The functional F_h is geodesically convex for W_γ if and only if h is convex on $\mathbb{R}^d$.*
2. *If in addition $h \in C^2(\mathbb{R}^d)$ and $\kappa \geq 0$, then F_h is κ -geodesically convex for W_γ if and only if*

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } z \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

Equivalently,

$$\xi^\top \nabla^2 h(z) \xi \geq \kappa \max_{Q \in \mathcal{K}_\gamma} \xi^\top Q \xi \quad \text{for every } z, \xi \in \mathbb{R}^d.$$

823 For $\kappa < 0$, this displayed scalar reformulation with a maximum is not equivalent to the
 824 matrix inequality; multiplication by the negative scalar reverses the extremum, so the
 825 corresponding pointwise scalar condition would involve $\min_{Q \in \mathcal{K}_\gamma} \xi^\top Q \xi$. No equivalence
 826 for negative κ is asserted here.

Proof. Assume first that h is convex and let

$$\mu_t = ((1-t)x + ty)_\# \pi_*$$

be any constant-speed W_γ -geodesic. Then

$$F_h(\mu_t) = \int h((1-t)x + ty) d\pi_*(x, y).$$

By the convexity of h ,

$$h((1-t)x + ty) \leq (1-t)h(x) + th(y),$$

827 and integrating yields geodesic convexity of F_h .

Conversely, if F_h is geodesically convex, apply the definition to Dirac masses $\mu_0 = \delta_x$ and $\mu_1 = \delta_y$. The unique geodesic is

$$\mu_t = \delta_{(1-t)x + ty},$$

so

$$h((1-t)x + ty) = F_h(\mu_t) \leq (1-t)h(x) + th(y),$$

828 which proves convexity of h .

Assume now that $h \in C^2$ and that

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } z \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

Let

$$\mu_t = ((1-t)x + ty)_\# \pi_*$$

be any constant-speed W_γ -geodesic, write $z_t = (1-t)x + ty$ and $\Delta = y - x$. Differentiating under the integral sign gives

$$\frac{d^2}{dt^2} F_h(\mu_t) = \int \Delta^\top \nabla^2 h(z_t) \Delta d\pi_*(x, y).$$

For every $Q \in \mathcal{K}_\gamma$ this implies

$$\frac{d^2}{dt^2} F_h(\mu_t) \geq \kappa \int \Delta^\top Q \Delta d\pi_*(x, y).$$

Since $\kappa \geq 0$, taking the maximum over $Q \in \mathcal{K}_\gamma$ yields

$$\frac{d^2}{dt^2} F_h(\mu_t) \geq \kappa \gamma \left(\int \Delta \Delta^\top d\pi_*(x, y) \right) = \kappa W_\gamma(\mu_0, \mu_1)^2.$$

829 Integrating twice in t gives the κ -geodesic convexity estimate.

Conversely, assume F_h is κ -geodesically convex. Testing the definition on the Dirac geodesic between δ_z and $\delta_{z+\xi}$ gives

$$h(z + t\xi) \leq (1-t)h(z) + th(z + \xi) - \frac{\kappa}{2} t(1-t) \gamma(\xi \xi^\top).$$

Differentiating at second order in t yields

$$\xi^\top \nabla^2 h(z) \xi \geq \kappa \gamma(\xi \xi^\top) = \kappa \max_{Q \in \mathcal{K}_\gamma} \xi^\top Q \xi.$$

Because $\kappa \geq 0$, this scalar inequality is equivalent to the matrix inequality

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } Q \in \mathcal{K}_\gamma.$$

830

□

831 The next remark interprets the condition of Theorem H.2 for the Schatten family.

Remark H.3 (Schatten norms). *For the Schatten geometries used throughout the paper and for $\kappa \geq 0$, the condition*

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } Q \in \mathcal{K}_\gamma$$

is equivalent to

$$\nabla^2 h(z) \succeq \kappa \text{Id}.$$

Indeed, for $p = 1$ we may choose $\mathcal{K}_\gamma = \{\text{Id}\}$. For $1 < p \leq \infty$, a convenient choice is

$$\mathcal{K}_\gamma = \{Q \succeq 0 : \|Q\|_{S_q} \leq 1\}, \quad q = \frac{p}{p-1},$$

832 *with the usual interpretation $q = 1$ when $p = \infty$. Under the standing condition $\kappa \geq 0$ in Theorem H.2,*
 833 *every such Q satisfies $Q \preceq \text{Id}$, and every rank-one projector belongs to $\mathcal{K}_\gamma$; hence the condition*
 834 *is equivalent to $\nabla^2 h(z) \succeq \kappa \text{Id}$, exactly as in the classical W_2 geometry. For negative κ , this*
 835 *simplification is not asserted.*

836 H.3 Relative Entropy

This subsection examines the geodesic convexity of relative entropy, which is a fundamental nonlinear example underlying diffusion-type gradient flows. Section H.4 treats in detail the case of a Gaussian target ν , for which Gaussians are stable, and boils down to the study of an ODE. Let

$$d\nu(x) = Z^{-1} e^{-V(x)} dx, \quad V \in C^2(\mathbb{R}^d),$$

and define the relative entropy

$$\text{Ent}_\nu(\mu) := \begin{cases} \int \log\left(\frac{d\mu}{d\nu}\right) d\mu, & \mu \ll \nu, \\ +\infty, & \text{otherwise.} \end{cases}$$

837 The trace case is the classical displacement-convexity theory of entropy. For general spectral
 838 geometries the same argument works only when the active quadratic costs remain uniformly elliptic,
 839 and otherwise one only gets a weaker statement by approximation.

840 The next theorem records the full all-geodesics result in the uniformly elliptic regime.

Theorem H.4 (Full entropy convexity under uniform ellipticity). *Assume γ is monotone and that $\mathcal{K}_\gamma$ can be chosen so that*

$$\mathcal{K}_\gamma \subset \mathbb{S}_{++}^d.$$

Assume moreover that

$$\nabla^2 V(x) \succeq \kappa Q \quad \text{for every } x \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

841 *Then Ent_ν is κ -geodesically convex on $(\mathcal{P}_2(\mathbb{R}^d), W_\gamma)$.*

Proof. Let

$$\mu_t = ((1-t)x + ty)_\# \pi_*$$

be any constant-speed W_γ -geodesic, and set

$$S_* = \int (y-x)(y-x)^\top d\pi_*(x, y).$$

Choose $Q_* \in \mathcal{K}_\gamma$ such that

$$\text{tr}(Q_* S_*) = \gamma(S_*) = W_\gamma(\mu_0, \mu_1)^2.$$

By Theorem 3.3, the same coupling π_* is optimal for the quadratic cost

$$c_{Q_*}(x, y) = (y-x)^\top Q_*(y-x),$$

and

$$W^{Q_*}(\mu_0, \mu_1)^2 = W_\gamma(\mu_0, \mu_1)^2.$$

Since Q_* is positive definite, let $L = Q_*^{1/2}$ and define

$$\tilde{\mu}_i = L_{\#}\mu_i, \quad \tilde{\nu} = L_{\#}\nu, \quad \tilde{\pi}_* = (L, L)_{\#}\pi_*.$$

Then $\tilde{\pi}_*$ is optimal for the Euclidean quadratic cost between $\tilde{\mu}_0$ and $\tilde{\mu}_1$, hence

$$\tilde{\mu}_t = ((1-t)z_0 + tz_1)_{\#}\tilde{\pi}_*$$

is a W_2 -geodesic and

$$W_2(\tilde{\mu}_0, \tilde{\mu}_1)^2 = W_{\gamma}(\mu_0, \mu_1)^2.$$

The reference measure $\tilde{\nu}$ has density

$$d\tilde{\nu}(z) = \tilde{Z}^{-1}e^{-\tilde{V}(z)} dz, \quad \tilde{V}(z) = V(L^{-1}z) + \text{constant},$$

so

$$\nabla^2 \tilde{V}(z) = L^{-1} \nabla^2 V(L^{-1}z) L^{-1}.$$

Because $Q_* \in \mathcal{K}_{\gamma}$, the curvature assumption gives

$$\nabla^2 V(x) \succeq \kappa Q_* = \kappa L^2,$$

hence

$$\nabla^2 \tilde{V}(z) \succeq \kappa \text{Id}.$$

The classical entropy convexity theorem for W_2 therefore yields

$$\text{Ent}_{\tilde{\nu}}(\tilde{\mu}_t) \leq (1-t) \text{Ent}_{\tilde{\nu}}(\tilde{\mu}_0) + t \text{Ent}_{\tilde{\nu}}(\tilde{\mu}_1) - \frac{\kappa}{2} t(1-t) W_2(\tilde{\mu}_0, \tilde{\mu}_1)^2.$$

Finally, relative entropy is invariant under the invertible change of variables L , so

$$\text{Ent}_{\tilde{\nu}}(\tilde{\mu}_t) = \text{Ent}_{\nu}(\mu_t),$$

842 and similarly at the endpoints. This gives the result. $\square$

843 The next theorem explains what remains true in the more relevant but singular regime where the
844 condition $\mathcal{K}_{\gamma} \subset \mathbb{S}_{++}^d$ fails. This lack of ellipticity is precisely what happens for Schatten- p geometries
845 with $p > 1$, because the corresponding representing sets contain singular matrices, but one still retains
846 a weaker geodesic-convexity statement by approximation.

Theorem H.5 (Weak entropy convexity). *Assume γ is monotone, that $\mathcal{K}_{\gamma} \subset \mathbb{S}_{+}^d$ is convex compact, and that*

$$\mathcal{K}_{\gamma} \cap \mathbb{S}_{++}^d \neq \emptyset.$$

Assume moreover that

$$\nabla^2 V(x) \succeq \kappa Q \quad \text{for every } x \in \mathbb{R}^d, Q \in \mathcal{K}_{\gamma}.$$

Then for every $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$ there exists at least one constant-speed W_{γ} -geodesic $(\mu_t)_{t \in [0,1]}$ from μ_0 to μ_1 such that

$$\text{Ent}_{\nu}(\mu_t) \leq (1-t) \text{Ent}_{\nu}(\mu_0) + t \text{Ent}_{\nu}(\mu_1) - \frac{\kappa}{2} t(1-t) W_{\gamma}(\mu_0, \mu_1)^2.$$

Proof. Choose $Q_0 \in \mathcal{K}_{\gamma} \cap \mathbb{S}_{++}^d$ and define

$$\mathcal{K}_{\gamma, \varepsilon} := (1-\varepsilon)\mathcal{K}_{\gamma} + \varepsilon Q_0, \quad \varepsilon \in (0, 1).$$

Then every element of $\mathcal{K}_{\gamma, \varepsilon}$ is positive definite. Let γ_{ε} be the support function of $\mathcal{K}_{\gamma, \varepsilon}$, namely

$$\gamma_{\varepsilon}(S) = \max_{Q \in \mathcal{K}_{\gamma, \varepsilon}} \text{tr}(QS) = (1-\varepsilon)\gamma(S) + \varepsilon \text{tr}(Q_0 S).$$

Hence

$$(1-\varepsilon)\gamma(S) \leq \gamma_{\varepsilon}(S) \leq \gamma(S)$$

for all $S \succeq 0$, and therefore

$$(1-\varepsilon)W_{\gamma}(\mu_0, \mu_1)^2 \leq W_{\gamma_{\varepsilon}}(\mu_0, \mu_1)^2 \leq W_{\gamma}(\mu_0, \mu_1)^2.$$

By Theorem H.4, each regularized geometry has full entropy convexity. Choose an optimal coupling π_ε for W_{γ_ε} and set

$$\mu_t^\varepsilon = ((1-t)x + ty)_\# \pi_\varepsilon.$$

Then

$$\text{Ent}_\nu(\mu_t^\varepsilon) \leq (1-t) \text{Ent}_\nu(\mu_0) + t \text{Ent}_\nu(\mu_1) - \frac{\kappa}{2} t(1-t) W_{\gamma_\varepsilon}(\mu_0, \mu_1)^2.$$

By compactness of couplings, one may extract a subsequence $\pi_{\varepsilon_n} \rightharpoonup \pi_*$. Setting

$$\mu_t = ((1-t)x + ty)_\# \pi_*,$$

the lower semicontinuity of relative entropy and the convergence

$$W_{\gamma_{\varepsilon_n}}(\mu_0, \mu_1)^2 \rightarrow W_\gamma(\mu_0, \mu_1)^2$$

847 yield the claimed inequality along the limit geodesic. The limit coupling π_* is W_γ -optimal by
848 continuity of the displacement covariance and of the support functions on trace-bounded sets. $\square$

849 The next proposition shows that, in the operator case, full geodesic convexity of relative entropy fails
850 in a very concrete way.

Proposition H.6 (Operator-norm counterexample). *Let $\gamma(S) = \lambda_{\max}(S)$ on $\mathbb{S}_+^2$, and let*

$$d\nu(x) = \frac{1}{2\pi} e^{-|x|^2/2} dx.$$

851 *Then Ent_ν is not geodesically convex on $(\mathcal{P}_2(\mathbb{R}^2), W_\gamma)$.*

Proof. Fix parameters $L > 0$ and $a > 0$ such that

$$\frac{4a^2}{3} \leq L^2.$$

Set

$$R_0 = [0, 1] \times [-a, a], \quad R_1 = [L, L+1] \times [-a, a],$$

and let

$$\mu_i = \frac{1}{2a} \mathbf{1}_{R_i} dx, \quad i = 0, 1.$$

Both endpoints are absolutely continuous with bounded support, hence satisfy

$$\text{Ent}_\nu(\mu_0) < +\infty, \quad \text{Ent}_\nu(\mu_1) < +\infty.$$

For any coupling π between μ_0 and μ_1 , define

$$S_\pi = \int (y - x)(y - x)^\top d\pi(x, y).$$

Then

$$\lambda_{\max}(S_\pi) \geq e_1^\top S_\pi e_1 = \int (y_1 - x_1)^2 d\pi(x, y) \geq \left(\int (y_1 - x_1) d\pi(x, y) \right)^2 = L^2.$$

852 Hence every admissible coupling has operator cost at least L^2 .

The translation map

$$T(x_1, x_2) = (x_1 + L, x_2)$$

pushes μ_0 to μ_1 and has covariance

$$S_T = \text{diag}(L^2, 0),$$

853 so $W_\gamma(\mu_0, \mu_1)^2 = L^2$.

Now consider the reflected transport

$$R(x_1, x_2) = (x_1 + L, -x_2).$$

It also pushes μ_0 to μ_1 because the second marginal of μ_0 is symmetric on $[-a, a]$. Its displacement is

$$R(x) - x = (L, -2x_2),$$

hence

$$S_R = \text{diag}\left(L^2, \frac{4a^2}{3}\right).$$

By the choice of a ,

$$\lambda_{\max}(S_R) = L^2,$$

854 so the coupling induced by R is again optimal for W_γ .

Consider the corresponding constant-speed geodesic

$$\mu_t = ((1-t)\text{Id} + tR)_\# \mu_0.$$

At time $t = \frac{1}{2}$ one gets

$$\mu_{1/2} = \left(x_1 + \frac{L}{2}, 0\right)_\# \mu_0,$$

which is the uniform measure on the horizontal segment

$$\left[\frac{L}{2}, \frac{L}{2} + 1\right] \times \{0\}.$$

Therefore $\mu_{1/2}$ is singular with respect to Lebesgue measure, hence also singular with respect to ν , and thus

$$\text{Ent}_\nu(\mu_{1/2}) = +\infty.$$

855 So along this optimal W_γ -geodesic the endpoints have finite entropy while the midpoint has infinite
856 entropy, which rules out geodesic convexity. $\square$

857 The next remark explains why Proposition H.6 does not contradict Theorem H.5.

858 **Remark H.7** (Why there is no contradiction). *Theorem H.5 is an existence statement: for each*
859 *pair of endpoints it guarantees at least one W_γ -geodesic along which the entropy inequality holds.*
860 *Proposition H.6 proves something different, namely that in the operator geometry there also exist*
861 *optimal geodesics along which the entropy blows up at an interior time. So the counterexample rules*
862 *out full all-geodesics convexity, but it does not preclude the existence of another optimal geodesic*
863 *selected by the approximation argument in Theorem H.5 and enjoying the convexity estimate.*

864 The following remark explains the scope of the entropy results for the Schatten family.

865 **Remark H.8** (Schatten norms). *The full theorem applies to the trace case $p = 1$, where one may*
866 *choose $\mathcal{K}_\gamma = \{\text{Id}\}$ and recover the classical W_2 displacement-convexity theory. For every Schatten*
867 *geometry with $1 < p \leq \infty$, the natural representing sets contain singular matrices, so the uniformly*
868 *elliptic argument above does not apply directly. Theorem H.5 nevertheless yields a weak geodesic-*
869 *convexity statement as soon as $\mathcal{K}_\gamma$ contains one positive definite matrix, which is the case for the*
870 *standard Schatten choices.*

871 H.4 Relative-entropy flows over Gaussians

872 The W_2 -gradient flow of the relative entropy $\text{Ent}_\nu(\mu) = \text{KL}(\mu | \nu)$ is the Langevin/Fokker–Planck
873 equation, whose stochastic single-particle realization is the overdamped Langevin diffusion. This
874 makes it a basic tool for sampling. We now describe what happens for the spectral Schatten geometries.
875 For $p > 1$, the resulting evolution is still explicit on Gaussian measures, but the PDE is nonlocal: its
876 velocity depends on global statistics of the current law, and therefore cannot be represented by an
877 autonomous single-particle SDE with coefficients depending only on the particle position.

878 **Proposition H.9** (Gaussian KL flow for Schatten geometries). *Let $\nu = \mathcal{N}(\bar{m}, \bar{\Sigma})$ with $\bar{\Sigma} \succ 0$, and*
879 *consider $\mathcal{F}(\mu) = \text{KL}(\mu | \nu)$. The Schatten- p flow solution μ_t of $\mathcal{F}$ is Gaussian $\mu_t = \mathcal{N}(m_t, \Sigma_t)$.*
880 *Define the classical $p = 1$ Gaussian velocity by $\dot{m}_t^{(1)} := -\bar{\Sigma}^{-1}(m_t - \bar{m})$ and $\dot{\Sigma}_t^{(1)} := 2I - \bar{\Sigma}^{-1}\Sigma_t -$
881 $\Sigma_t\bar{\Sigma}^{-1}$. Then the trace geometry $p = 1$ satisfies $\dot{m}_t = \dot{m}_t^{(1)}$ and $\dot{\Sigma}_t = \dot{\Sigma}_t^{(1)}$.*

882 For a general Schatten geometry $p \in [1, \infty]$, set $q := 2p/(2p - 1)$, with the convention $q = 1$ when
 883 $p = \infty$, and define

$$F_t := (\bar{\Sigma}^{-1} - \Sigma_t^{-1})\Sigma_t(\bar{\Sigma}^{-1} - \Sigma_t^{-1}) + \bar{\Sigma}^{-1}(m_t - \bar{m})(m_t - \bar{m})^\top \bar{\Sigma}^{-1}.$$

884 Assume $F_t \succ 0$ and define

$$\mathcal{N}_t^q := \text{tr}(F_t^{q/2}), \quad P_{p,t} := \mathcal{N}_t^{2-q} F_t^{q/2-1}.$$

885 Then the Schatten- p Gaussian KL flow is

$$\dot{m}_t = P_{p,t} \dot{m}_t^{(1)}, \quad \dot{\Sigma}_t = \frac{1}{2} \left(P_{p,t} \dot{\Sigma}_t^{(1)} + \dot{\Sigma}_t^{(1)} P_{p,t} \right).$$

886 In particular, for $p = 1$, one has $q = 2$, $P_{1,t} = I$, and the classical Langevin covariance equation is
 887 recovered. For $p = \infty$, one has

$$P_{\infty,t} = \text{tr}(F_t^{1/2}) F_t^{-1/2}.$$

888 *Proof.* For $\mu_t = \mathcal{N}(m_t, \Sigma_t)$ and $\nu = \mathcal{N}(\bar{m}, \bar{\Sigma})$, the first variation of $\text{KL}(\mu | \nu)$ has Wasserstein
 889 gradient

$$g_t(x) = \nabla_x \log \frac{d\mu_t}{d\nu}(x) = (\bar{\Sigma}^{-1} - \Sigma_t^{-1})(x - m_t) + \bar{\Sigma}^{-1}(m_t - \bar{m}).$$

890 Thus g_t is affine. Set $H_t := \bar{\Sigma}^{-1} - \Sigma_t^{-1}$ and $b_t := \bar{\Sigma}^{-1}(m_t - \bar{m})$, so that $g_t(x) = H_t(x - m_t) + b_t$.
 891 Since $\int (x - m_t) d\mu_t(x) = 0$ and $\int (x - m_t)(x - m_t)^\top d\mu_t(x) = \Sigma_t$, the covariance matrix of g_t is
 892 exactly

$$\int g_t(x) g_t(x)^\top d\mu_t(x) = H_t \Sigma_t H_t + b_t b_t^\top = F_t.$$

893 For the Schatten geometry associated with p , the Measure-LMO gives the velocity

$$v_t(x) = -P_{p,t} g_t(x), \quad P_{p,t} = \mathcal{N}_t^{2-q} F_t^{q/2-1}, \quad \mathcal{N}_t^q = \text{tr}(F_t^{q/2}).$$

894 This formula is the Schatten duality map applied to the vector field g_t . Because g_t is affine and $P_{p,t}$ is
 895 independent of x , the velocity v_t is affine in x . Therefore the continuity equation $\partial_t \mu_t + \text{div}(\mu_t v_t) = 0$
 896 preserves the Gaussian family.

897 We now compute the induced ODE on mean and covariance. Since $v_t(x) = -P_{p,t} H_t(x - m_t) - P_{p,t} b_t$,
 898 the mean satisfies

$$\dot{m}_t = \int v_t(x) d\mu_t(x) = -P_{p,t} b_t.$$

899 But $b_t = \bar{\Sigma}^{-1}(m_t - \bar{m})$, hence $-b_t = \dot{m}_t^{(1)}$. Therefore $\dot{m}_t = P_{p,t} \dot{m}_t^{(1)}$.

900 For the covariance, using $\dot{\Sigma}_t = \int (v_t(x)(x - m_t)^\top + (x - m_t)v_t(x)^\top) d\mu_t(x)$, we obtain

$$\dot{\Sigma}_t = -P_{p,t} H_t \Sigma_t - \Sigma_t H_t P_{p,t}.$$

901 On the other hand,

$$\dot{\Sigma}_t^{(1)} = 2I - \bar{\Sigma}^{-1} \Sigma_t - \Sigma_t \bar{\Sigma}^{-1}.$$

902 Since $H_t = \bar{\Sigma}^{-1} - \Sigma_t^{-1}$ and both $\bar{\Sigma}^{-1}$ and Σ_t^{-1} are symmetric,

$$-H_t \Sigma_t = I - \bar{\Sigma}^{-1} \Sigma_t, \quad -\Sigma_t H_t = I - \Sigma_t \bar{\Sigma}^{-1}.$$

903 Therefore

$$-P_{p,t} H_t \Sigma_t - \Sigma_t H_t P_{p,t} = P_{p,t} (I - \bar{\Sigma}^{-1} \Sigma_t) + (I - \Sigma_t \bar{\Sigma}^{-1}) P_{p,t}.$$

904 When $P_{p,t}$ commutes with the symmetric part of the classical covariance velocity, this is exactly
 905 $\frac{1}{2}(P_{p,t} \dot{\Sigma}_t^{(1)} + \dot{\Sigma}_t^{(1)} P_{p,t})$. More generally, the correct intrinsic twisted form is

$$\dot{\Sigma}_t = P_{p,t} (I - \bar{\Sigma}^{-1} \Sigma_t) + (I - \Sigma_t \bar{\Sigma}^{-1}) P_{p,t}.$$

906 In the common commuting setting, or whenever $\bar{\Sigma}^{-1} \Sigma_t = \Sigma_t \bar{\Sigma}^{-1}$, this reduces to the displayed
 907 symmetrized formula in the statement.

908 Finally, for $p = 1$ one has $q = 2$, hence $\mathcal{N}_t^{2-q} = 1$ and $F_t^{q/2-1} = I$, so $P_{1,t} = I$. The equations
 909 therefore reduce to $\dot{m}_t = -\bar{\Sigma}^{-1}(m_t - \bar{m})$ and $\dot{\Sigma}_t = 2I - \bar{\Sigma}^{-1} \Sigma_t - \Sigma_t \bar{\Sigma}^{-1}$, which is the Gaussian
 910 restriction of the classical Langevin/Fokker-Planck flow. For $p = \infty$, $q = 1$, giving $P_{\infty,t} =$
 911 $\text{tr}(F_t^{1/2}) F_t^{-1/2}$. $\square$

912 I Positively 2-Homogeneous Case and a Generalized Unbalanced Geometry

This section continues the two-layer mean-field discussion of Section 5 in the positively two-homogeneous case. The point is that such models reduce the ambient transport problem to an unbalanced transport problem on the sphere, and this section explains how the spherical dynamics inherits a spectral action from the ambient flow. Let

$$\mathcal{F}(\mu) = R\left(\int_{\mathbb{R}^d} \Phi(x) d\mu(x)\right), \quad \Phi(\lambda x) = \lambda^2 \Phi(x).$$

Writing $x = r\omega$ with $\omega \in \mathbb{S}^{d-1}$, define the weighted spherical projection

$$\int_{\mathbb{S}^{d-1}} \psi(\omega) d\Pi_2(\mu)(\omega) := \int_{\mathbb{R}^d} |x|^2 \psi\left(\frac{x}{|x|}\right) d\mu(x).$$

913 The first point is that two-homogeneous models only depend on this weighted spherical projection.

Proposition I.1 (Exact quotient). *There exists a functional $\bar{f}$ on $\mathcal{M}_+(\mathbb{S}^{d-1})$ such that*

$$\mathcal{F}(\mu) = \bar{f}(\Pi_2(\mu)).$$

Proof. The homogeneity identity $\Phi(r\omega) = r^2 \Phi(\omega)$ immediately gives

$$\int_{\mathbb{R}^d} \Phi(x) d\mu(x) = \int_{\mathbb{S}^{d-1}} \Phi(\omega) d\Pi_2(\mu)(\omega),$$

914 which defines $\bar{f}$. □

The second point is that the ambient normalized velocity splits into radial and tangential parts on the sphere. If a vector field v is 1-homogeneous, namely $v(\lambda x) = \lambda v(x)$ for every $\lambda > 0$, then for every $x = r\omega \neq 0$ it can be written uniquely as

$$v(r\omega) = r(b(\omega)\omega + \tau(\omega)), \quad \tau(\omega) \in T_\omega \mathbb{S}^{d-1}.$$

Indeed, one defines

$$b(\omega) := v(\omega) \cdot \omega, \quad \tau(\omega) := v(\omega) - (v(\omega) \cdot \omega)\omega,$$

so that $\tau(\omega) \cdot \omega = 0$, and then 1-homogeneity gives

$$v(r\omega) = r v(\omega) = r(b(\omega)\omega + \tau(\omega)).$$

915 The decomposition is unique because it is simply the orthogonal splitting of $v(\omega)$ into its normal and
 916 tangential parts on the sphere. In the positively two-homogeneous setting considered here, the first
 917 variation is 2-homogeneous and its spatial gradient is therefore 1-homogeneous; correspondingly, the
 918 natural steepest-descent velocity fields for the flow belong to this 1-homogeneous class. The next
 919 proposition computes the projected spherical PDE.

Proposition I.2 (Projected continuity-reaction equation). *If μ_t solves the Spectral Wasserstein flow and $\nu_t = \Pi_2(\mu_t)$, then*

$$\partial_t \nu_t + \operatorname{div}_{\mathbb{S}^{d-1}}(\nu_t \tau_t) = 2b_t \nu_t.$$

920 *Proof.* Test the ambient continuity equation against the lifted observable $\tilde{\psi}(r\omega) = r^2 \psi(\omega)$ and
 921 identify the radial and tangential contributions. □

Motivated by Proposition I.2, define for nonnegative measures ν_0, ν_1 on $\mathbb{S}^{d-1}$

$$\operatorname{UW}_\gamma(\nu_0, \nu_1)^2 := \inf_{(\nu_t, b_t, \tau_t)} \int_0^1 \gamma\left(\int_{\mathbb{S}^{d-1}} (b_t(\omega)\omega + \tau_t(\omega))(b_t(\omega)\omega + \tau_t(\omega))^\top d\nu_t(\omega)\right) dt,$$

under the continuity-reaction constraint

$$\partial_t \nu_t + \operatorname{div}_{\mathbb{S}^{d-1}}(\nu_t \tau_t) = 2b_t \nu_t.$$

922 The next proposition identifies the ambient homogeneous action with this spherical unbalanced
 923 action.

924 **Proposition I.3** (Ambient action equals spherical action). *For positively two-homogeneous models*
925 *and 1-homogeneous velocities, the ambient Spectral Wasserstein action equals the spherical action*
926 *defining UW_γ .*

Proof. Write $x = r\omega$ and $v(r\omega) = r(b(\omega)\omega + \tau(\omega))$. By definition of $\Pi_2(\mu)$,

$$\int_{\mathbb{R}^d} v(x)v(x)^\top d\mu(x) = \int_{\mathbb{S}^{d-1}} (b(\omega)\omega + \tau(\omega))(b(\omega)\omega + \tau(\omega))^\top d\Pi_2(\mu)(\omega).$$

927 Inserting this identity into the Benamou–Brenier action yields the claim. $\square$

928 The following remark explains how the trace-norm case collapses to the classical Wasserstein–Fisher–
929 Rao geometry.

Remark I.4. *When $\gamma(S) = \text{tr}(S)$, the integrand becomes*

$$\int_{\mathbb{S}^{d-1}} (|\tau_t(\omega)|^2 + b_t(\omega)^2) d\nu_t(\omega).$$

930 *Since the reaction rate is $\alpha_t = 2b_t$, this is exactly the classical Wasserstein–Fisher–Rao action used*
931 *by Chizat and Bach [2018] for positively homogeneous ReLU-type mean-field models. In the trace*
932 *case, this dynamic spherical geometry has static formulations equivalent to the Wasserstein–Fisher–*
933 *Rao or Hellinger–Kantorovich distances [Chizat et al., 2018, Liero et al., 2018]. For general γ , the*
934 *construction above gives a generalized unbalanced transport geometry on the sphere, now available*
935 *for every Schatten endpoint and interpolation. Beyond the $p = 1$ trace case, it is not clear whether the*
936 *resulting UW_γ admits an analogous static formulation; understanding this would be an interesting*
937 *direction for future work.*

938 J Notation

939 This section collects the main notation used throughout the paper. The first table lists standard
940 background notation, while the second table lists the notation introduced for the Spectral Wasserstein
941 geometry.

Table 1: Classical notation.

Notation	Name	First use
$\mathbb{R}^d$	Euclidean space	Sec. 2
$\mathbb{S}^d$	Symmetric matrices	App. A
$\mathbb{S}_+^d$	PSD matrices	Sec. 2
$\mathbb{S}_{++}^d$	Positive definite matrices	Thm. 2.5
$S \preceq T$	Loewner order	Sec. 2
I, Id	Identity matrix/map	Sec. 2
$\text{tr}(S)$	Trace	Def. 2.1
$\text{diag}(a_i)$	Diagonal matrix	Sec. 2
$\lambda(S)$	Eigenvalues of S	Sec. 2
$\ \cdot\ _F$	Frobenius norm	Def. 2.2
$\langle \cdot, \cdot \rangle_F$	Frobenius product	Def. 2.2
$\ A\ _{S_p}$	Schatten p -norm	Sec. 2
$A^\dagger$	Pseudoinverse	Eq. (3)
$\mathcal{P}_2(\mathbb{R}^d)$	Measures with finite second moment	Def. 2.3
δ_x	Dirac mass	Sec. 2
$T_\# \mu$	Pushforward	Def. C.1
$\Pi(\mu, \nu)$	Couplings	Def. 3.1
$L^2(\mu; \mathbb{R}^d)$	Square-integrable vector fields	Def. 2.3
∇, ∇_x	Euclidean gradient	Sec. 2
div	Divergence	Eq. (2)
$\delta\mathcal{F}/\delta\mu$	First variation	Sec. 2
W_2	Quadratic Wasserstein distance	Sec. 1

Notation	Name	First use
$\mathcal{N}(m, \Sigma)$	Gaussian law	Prop. 3.4
Σ	Covariance matrix	Prop. 3.4
$\mathbb{S}^{d-1}$	Unit sphere	App. I
$\nabla_{\mathbb{S}^{d-1}}, \text{div}_{\mathbb{S}^{d-1}}$	Spherical gradient/divergence	Prop. I.2

Table 2: Main notation introduced in the paper.

Notation	Name	Introduction
γ	Matrix norm on $\mathbb{S}_+^d$	Sec. 2
γ_p	Schatten geometry	Sec. 2
p, q	Primal/dual exponents	Rem. A.4
$\mathcal{K}_\gamma$	Representing set	Def. 2.1
$\mathcal{P}_\gamma$	Polar set	App. A
$\mathcal{K}_{\gamma_p}$	Schatten representing set	Rem. A.4
$N(V)$	Matrix tangent norm	Sec. 2
$J_\gamma(G)$	Matrix LMO	Def. 2.2
$J_p(G)$	Schatten matrix LMO	Prop. A.5
X	Particle matrix	Sec. 2
μ_X	Empirical particle measure	Sec. 2
$F(X)$	Particle objective	Sec. 2
$\mathcal{F}(\mu)$	Mean-field objective	Sec. 2
$\mathcal{N}_\mu(v)$	Measure tangent norm	Def. 2.3
$\mathcal{J}_\gamma^\mu(g)$	Measure-LMO	Def. 2.3
$S_\mu(g)$	Gradient covariance	Thm. 2.5
Q_μ^*	Inverse-trace minimizer	Thm. 2.5
g_μ	Wasserstein gradient	Sec. 2
W_γ	Static Spectral Wasserstein distance	Def. 3.1
W^Q	Fixed-metric transport cost	Thm. 3.3
c_γ, C_γ	Trace comparison constants	Prop. 3.2
W_γ^{BB}	Dynamic Spectral Wasserstein distance	Def. 4.1
$\mathcal{A}_\gamma(\mu, m)$	Momentum action	Prop. E.1
W_γ^{M}	Monge-restricted cost	Def. C.1
$\phi(z, x)$	Feature map	Sec. 5
H_μ	Mean-field predictor	Sec. 5
$R(H)$	Risk functional	Sec. 5
$\text{MMD}(\mu, \nu)^2$	Squared MMD	Sec. 5
(Q, K, V, O)	Attention weights	Sec. 5
N_{eff}	Effective tokens	Sec. 5
C_ρ	Data covariance	Sec. 5
B_ρ	Teacher cross-moment	Sec. 5
M_μ	Learned linear predictor	App. F
$A_\star$	Teacher matrix	App. F
E_t	Linear prediction error	Prop. F.2
$r_i(t)$	Modal predictor coefficient	Prop. 5.3
$s_i(t)$	Modal variance	Prop. 5.3
β_i	Teacher mode	Prop. 5.3
$\mathcal{N}(r, s)$	Modal normalization	Prop. 5.3
$\mathbb{I}_p(V)$	Momentum constitutive map	Eq. (4)
$F_h(\mu)$	Linear measure functional	Thm. H.2
κ	Geodesic-convexity constant	Def. H.1
$\text{Ent}_\nu(\mu)$	Relative entropy	App. H
Φ	Two-homogeneous feature map	App. I
$\Pi_2(\mu)$	Spherical projection	App. I
UW_γ	Unbalanced spherical action	App. I